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GRADUATION PROJECT

NATIONAL ENGINEERING DEGREE

SPECIALTY : INFORMATION TECHNOLOGY

TITLE : Enhancing an E-Recruitment Platform with
Advanced Profile Management and a Dynamic
Customizable CV Design and Generation Tool

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Abstract

Abstract :

This report is the result of an end-of-study project focused on the development of advanced features for an e-recruitment platform, HireCue. The project aimed to enhance both candidate and recruiter experiences through LinkedIn data extraction, AI-driven onboarding, and a dynamic profile management system for candidates, alongside a highly interactive template library, a feature-rich PDF designer, and custom CV plugins for recruiters. The solution provides seamless profile management and powerful document creation tools tailored to recruitment needs.

Key Words : E-recruitment, PDF Designer, CV Plugins, AI-Driven Onboarding, MERN Stack, JavaScript, Node.js, React.js, MySQL, Sequelize ORM.

Résumé :

Ce rapport est le résultat d'un projet de fin d'études axé sur le développement de fonctionnalités avancées pour une plateforme de recrutement en ligne, *HireCue*. Le projet vise à améliorer l'expérience des candidats grâce à l'extraction de données LinkedIn, à un formulaire d'intégration alimenté par l'IA, et à une gestion dynamique des profils. Pour les recruteurs, il offre une bibliothèque de modèles interactive, un concepteur PDF riche en fonctionnalités, et des plugins personnalisés pour les CV. La solution garantit une gestion fluide des profils et des outils de création de documents adaptés aux besoins du recrutement.

Mots clés : Recrutement en ligne, Concepteur PDF, Plugins CV, Intégration IA, MERN Stack, JavaScript, Node.js, React.js, MySQL, Sequelize ORM.

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General Introduction

In today's fast-paced world, e-recruitment has become an indispensable part of the modern talent acquisition process. One of the key challenges in the recruitment process is ensuring that candidates can effectively manage and present their qualifications while recruiters can efficiently evaluate them. To address this, our platform facilitates the candidate onboarding and profile management process, enabling applicants to maintain accurate and consistent profiles. Additionally, recruiters often receive CVs in various formats, which complicates the comparison and evaluation of candidates. To streamline this process, we implemented a CV designer and generator tool that standardizes the format of submitted CVs, helping recruiters save time and improve the efficiency of candidate assessments.

What are the best ways to streamline candidate profile management? How can we ensure that recruiters receive CVs in a standardized format for easy comparison? How can we improve the overall efficiency of the recruitment process?

The main objective of this project is to enhance the recruitment platform by developing a streamlined onboarding and profile management system for candidates and implementing a CV designer and generator tool for recruiters. This report is divided into two main parts and five sprints, as well as a general introduction and conclusion, to present the work completed during this internship. The first chapter, entitled "General Presentation Of The Project," introduces the project and its environment, starting with a presentation of the host company, Teligencia, and its field of activity. We then present the existing situation and the problem, the work required, and the specifications. Finally, we outline the methodology adopted in our project. The second chapter, "Preparation Of The Project," covers the identification of stakeholders and their roles, the specification of functional and non-functional requirements, the product backlog, the project plan, and the selection of architectural and technical requirements. Finally, we dedicate two chapters to the two main releases of the project, where each chapter introduces each sprint with its sprint backlog, modeling, and implementation through graphical interfaces.

CHAPTER 1

General Presentation Of The Project

1.1 Introduction

First chapter is a general introduction to the project. First, we will present the hosting company. Then, we will introduce our project and we discuss some of the similar project to our own and the methodology that will be use during the realisation of this project.

1.2 Presentation of The Hosting Company

Before proceeding, it is essential to identify the various elements of the environment. Therefore, we will start with a presentation of the hosting company.

1.2.1 Company Profile

Teligencia is a company that specialises mainly in cyber security. It provides its clients with an appropriate security plan to protect their business from any kind of cyber threats. In addition, it is also specialised in the development of IOT and AI. It offers several artificial intelligence-based solutions. It was established in the year 2020, in Germany. Moreover, it has 50 employees [1].

1.2.2 Branches of Teligencia

Teligencia is an international company. Its headquarter is located in Düsseldorf, Germany. Besides its headquarter office, it has two branches : the first one is located in Karlsfeld, Germany and the other one is located in Tunis, Tunisia.



Figure 1.1 – Branches of Teligencia [1].

1.2.3 Teligencia's Services

Teligencia provides a multitude of services, including [14] :

- **Infrastructure Penetration Testing** : This service involves the identification of vulnerabilities within the network and the implementation of measures to ensure the security of the system.

- **Hardware Device Penetration Testing** :The objective of this testing is to identify any potential vulnerabilities in the sensors and hardware products under consideration.
- **Social Engineering Simulation** :Social engineering is the practice of employing deceiving techniques with the intention of manipulating individuals into giving sensitive information.
- **Audit and Standards implementation** :The service offers an audit of an organisation against international information security standards, such as ISO 27001.
- **Threat Intelligence and public exposure Analysis** :It evaluate the level of exposition of the IT services provided by the company and its employees.
- **Consultancy and Advisory** :This service consists of providing the adequate security solutions and strategies that fit the customer's business.
- **Secure Development life Cycle** :Enhance your development lifecycle to be SSDLC (Secure Software Development Lifecycle) compliant.
- **Web penetration testing** : The service is based on the detection of the cybersecurity risk of the web application.
- **Vulnerability Management Services** : Vulnerability management services are designed to identify cybersecurity threats and recommend areas for improvement across the organization's networks, applications and devices.
- **Products Audits and After sales services** : The integration of digital workflows and the retention of control over customer data, independently of suppliers.

1.3 Presentation of the Project

In this section, we will present the current solution developed by the company and the new features to be added. We will also examine existing solutions that offer similar features.

1.3.1 The Current Solution "HireCue":

HireCue is an e-recruitment platform developed by Teligencia. This platform provides all the necessary functionalities required for the recruitment process, including the ability to post new job offers, submit applications...

Figure 1.2 represents the various functionalities offered by HireCue.

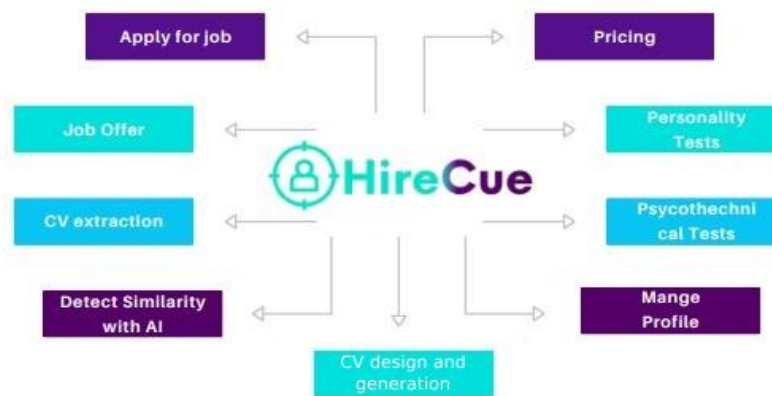


Figure 1.2 – HireCue’s Features.

1.3.2 Project’s Problematic

The platform allows candidates to apply for job offers and manage their profiles. However, candidates often face challenges in maintaining consistent and updated information across multiple applications, which can lead to incomplete or inaccurate profiles. Additionally, recruiters receive CVs in various formats, making it difficult to compare and evaluate candidates effectively. To address these issues, we proposed the creation of a streamlined onboarding and profile management system for candidates. Furthermore, to assist recruiters in the evaluation process, we implemented a CV designer and generator tool that standardizes the format of submitted CVs.

1.3.3 Study of Similar Solutions

In this section, we examine some of the existing solutions available on the market that share similar features with those we aimed to implement in our project. These solutions were selected based on their popularity and relevance to candidate profile management and CV standardization.

LinkedIn

LinkedIn is a widely used platform for professional networking and job searching. It allows candidates to create detailed profiles, including their work experience, education, skills, and more. LinkedIn provides recruiters with a standardized view of candidate profiles, making it easier to compare qualifications. Additionally, LinkedIn offers tools for CV generation and profile export, which help standardize candidate information for recruiters during the hiring process.

Jobvite

Jobvite is an applicant tracking system (ATS) that includes a robust candidate profile management feature. It allows candidates to input and manage their information in a

structured format, which recruiters can easily access and evaluate. Jobvite also offers tools to generate and export candidate profiles into standardized formats, facilitating the comparison of candidates' qualifications and making the recruitment process more efficient.

Zety

Zety is an online CV builder that helps candidates create professional CVs using customizable templates. The platform guides users through the CV creation process, ensuring that all necessary information is included and formatted consistently. For recruiters, Zety provides the benefit of receiving CVs in a standardized format, which simplifies the evaluation process and helps identify the best candidates more quickly.

1.3.4 Synthesis

Part 1: Onboarding and Profile Management

Once we presented the various solutions for onboarding and profile management, we conducted a comparative analysis of the platforms based on their features and effectiveness. The results are summarized in Table 1.1.

Table 1.1 – Comparative study of onboarding and profile management platforms.

Criteria	LinkedIn	Jobvite	Zety
Onboarding Process	Yes	Yes	No
Profile Management	Yes	Yes	No
Customization for Recruiters	No	Yes	No
Automated Profile Updates	No	Yes	No
User-Friendliness	High	Moderate	High
Multi-language Support	No	Yes	No
Mobile Accessibility	Yes	Yes	No
Automatic Data Import from LinkedIn	Yes		No
CV Upload and Data Extraction	No	Yes	No

Our analysis revealed that LinkedIn and Jobvite excel in onboarding and profile management, with Jobvite offering greater customization options and integration with other HR systems. LinkedIn supports automatic data import from LinkedIn profiles, while Jobvite allows CV uploads to extract data. However, neither provides automated profile updates or multi-language support. Zety is user-friendly and offers excellent CV creation tools but does not address onboarding or profile management needs. This indicates a requirement for a solution that combines onboarding, profile management, and

customization with additional features such as automated data import and CV data extraction.

Part 2: CV Design and Generation

To evaluate CV design and generation solutions, we reviewed several platforms based on their features and capabilities. The comparative results are detailed in Table 1.2.

Table 1.2 – Comparative study of CV design and generation platforms

Criteria	TestGorilla	CodinGame	TogglHire
Randomized questions per test	yes	yes	yes
Set time limits	yes	yes	yes
Disable Copy and Paste	yes	yes	no
IP address detection	yes	yes	yes
Location detection	yes	yes	no
Detect device's features	yes	no	no
Detect if the candidate left the assessment window	yes	yes	no
Prevent multiple attempts	yes	yes	yes
Take snapshots	yes	no	no
Take screenshots	no	no	no
Customising the anti-cheating parameters	no	no	no

The comparison shows that while LinkedIn and Jobvite offer CV export features and some level of template customization, they fall short in terms of bulk CV generation and real-time updates. Zety provides extensive template customization but does not integrate with recruitment systems or offer automatic CV generation for all candidates. Additionally, LinkedIn supports automatic data import from LinkedIn profiles, while Jobvite allows CV uploads to extract data. This analysis underscores the need for a solution that facilitates effective CV design and generation, integrates with recruitment workflows, supports bulk generation, and includes capabilities for importing and extracting data from LinkedIn profiles or uploaded CVs.

Part 3: Conclusion

From the comparative analysis of onboarding, profile management, and CV design platforms, it is evident that no single solution currently offers a comprehensive set of features that meets all the needs identified. The ideal solution should integrate

effective onboarding and profile management, support extensive CV customization, provide automated features for bulk CV generation, and include capabilities for importing and extracting data from LinkedIn profiles or uploaded CVs. This would enhance efficiency in recruitment processes and better meet the needs of both recruiters and candidates.

1.3.5 Proposed Solution

This project aims to significantly enhance the HireCue platform by introducing new functionalities that streamline candidate management and improve the recruitment process. The proposed solution is divided into two main components:

Part 1: Automated CV Generation and Data Import

To facilitate a more efficient hiring process, the following features will be implemented:

- **Automated CV Generation:** The system will automatically generate CVs for all candidates associated with a particular job, eliminating the need for manual one-by-one generation.
- **Data Import from LinkedIn:** Candidates can import their professional data directly from LinkedIn, streamlining the application process and ensuring up-to-date information.
- **CV Data Extraction:** The platform will allow candidates to upload their existing CVs, from which relevant data will be automatically extracted and populated into the system.

Part 2: Enhanced Recruitment Workflow

To further improve the candidate management process, the following functionalities will be introduced:

- **Bulk Candidate Management:** Recruiters will be able to manage multiple candidates simultaneously, enabling more efficient handling of large applicant pools.
- **Customizable CV Templates:** Recruiters can choose from various CV templates to automatically format the extracted or generated candidate data, ensuring consistency and professionalism in presentation.
- **Integrated Communication Tools:** The platform will include tools for automated emails and notifications, keeping candidates informed about their application status without manual intervention.

These enhancements will streamline the recruitment process, reduce manual work for recruiters, and provide a more seamless experience for candidates.

1.4 Choice of Working Method

A software development process is a set of activities that lead to the production of software or the enhancement of an existing system. Development methodologies aim to help facilitate project management and develop a solution that meets requirements, is maintainable and well documented.

1.4.1 Agile Methodologies

The agile methodology is a process that makes it easier for teams to react smoothly and avoid unpredictable reactions. It creates numerous evaluation units to assess where a project stands during the development period. Teams evaluate the project in regular meetings called sprints or iterations. In order to choose the most appropriate agile method for our application, it is essential to compare between some of the most used agile methods. In the table 1.3, we have conducted a comparative study between some of the agile methodologies which are : XP (Extreme Programming), Kanban, Scrum.

Table 1.3 – Comparative study of agile methodologies.

Methodology	Advantages	Disadvantages
XP (Extreme Programming)	— Ensure a fast paced working environment	— Focus on coding rather than designing
	with no time wasted in the process — Easy to implement — Guarantee of software stability thanks to continuous testing	— Lack of documentation.
Kanban	— simplicity. — Continuous throughout the project. — Encourage team work.	— There are no team roles. — Not defined by time frames.

Scrum	— Scrum ensures efficient use of time and money. — Developments are programmed and reviewed during the sprint. — Short sprints make it easier to make changes based on feedback much more easily. — Large projects are broken down into manageable sprints.	— The risk of project failure is high if people are not very determined or too cooperative. — Restart after each sprint.
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1.4.2 Chosen Methodology

Therefore, our choice of methodology for this project is Scrum. Scrum is a highly effective project management framework that promotes collaboration, adaptability, and efficiency in complex tasks. Its iterative approach allows teams to respond quickly to changing requirements, fostering continuous improvement and delivering valuable, high-quality products.

Scrum has mainly five phases which are :

- Initiation
- Planning and Estimation
- Implementation
- Reviewing
- Releasing

1.5 Conclusion

This chapter allowed us to make a general presentation of the project, starting with a presentation of the host company, then a description of the existing solutions and their limitations. Then we detailed the objectives we wanted to achieve. Finally, we discussed and chose the most appropriate methodology. In the next chapter we will describe in detail the preparation phase of our project.

CHAPTER 2

Requirements Analysis and Specification

2.1 Introduction

The project preparation phase is undoubtedly a critical stage in the project cycle. In this chapter, we start by identifying the functional and non-functional requirements and the actors in our system. This analysis will allow us to assimilate a product backlog and ensure that it is planned in sprints. Finally, we conclude this chapter with a presentation of the architectural and technological choices required to complete this project.

2.2 Requirements Specification

Identifying requirements is an essential stage in the software development cycle. In fact, specifying the actors and functionalities ensures that the developer creates an application that meets the customer's needs.

2.2.1 Actors Identification

The identification of the actors in the application is important for the fulfilment of their needs. Therefore, this section is an introduction to the five actors in the application :

- **Company Admin** : This role provides high-level access to companies using our SaaS platform, allowing them to effectively manage the recruiters and candidates within their organization. Company Administrators are responsible for adding, modifying and monitoring recruiter and candidate accounts within their company, as well as managing job postings specific to their organization. However, their access is limited to their company level and they do not have global administrator permissions.
- **Recruiter** : This role is assigned to members of the company who are responsible for managing job opportunities. Recruiters can be associated with a company or they can also be individuals. They are responsible for the management, the consultation, the addition, the modification and the deletion of the job offers published on the site. They can also be in charge of the interaction with the applicants.
- **Candidate** : This role represents the jobseekers. They have access to all the general functions available to candidates. They can search for jobs and apply for vacancies.

2.2.2 Functional Requirements

Functional requirements represent the expectations of the target actor. The following functionalities must be fulfilled by our application :

- ❖ **Company Admin, Recruiter** : can :
 - **Job and Candidate Management:**

- Create, update, and delete job profiles.
- View all candidates associated with a specific job profile.
- Automatically generate CVs for all candidates related to a specific job profile.

➤ **CV Management:**

- Access and use a CV editor to create and customize CV templates.
- Generate CVs for candidates using predefined or customized templates.
- Automatically generate CVs for all candidates related to a specific job profile without manually processing each candidate.

➤ **Skills Management:**

- Add new skills.
- Delete skills.
- View all skills.
- Search for skills based on their name.

❖ **Candidate : can :**

➤ **Onboarding:**

- Upload their CV to have their data automatically extracted.
- Import data directly from their LinkedIn profile during onboarding.
- Manually fill in their personal data during the onboarding process.

➤ **Profile Management:**

- Check and edit all personal data and preferences in the profile settings page.
- Update personal information and preferences through the profile settings page.

2.2.3 Non Functional Requirements

Non-functional requirements represent the rules to be followed to guarantee the quality of the application to be completed.

➤ **Integrity :**

- It's crucial to ensure the integrity of the application, especially in candidate data management and CV generation. Measures should be taken to prevent any unauthorized manipulation of candidate information.

➤ **Security :**

- Protecting user data is paramount. We need to ensure secure connections using the HTTPS protocol and implement a reverse proxy with NGINX to safeguard data transmission.
- All data uploaded or imported (e.g., from LinkedIn or CVs) must be securely processed and stored to prevent data breaches.

➤ **Authorisation :**

- Role-based access control is essential to ensure that only authorized users (e.g., recruiters, company admins) can access sensitive functionalities like CV editing and generation, while candidates can access onboarding and profile management features.
- Each role will have clearly defined permissions to restrict access to functionalities not relevant to their role.

➤ **Ergonomics :**

- The user interface should be intuitive and user-friendly, enabling recruiters to easily manage candidate profiles, generate CVs, and utilize the CV editor.
- For candidates, the onboarding process should be seamless, whether they choose to upload a CV, import data from LinkedIn, or manually fill in their information. The profile settings page should be easy to navigate for managing personal data and preferences.

2.3 Global Use Case Diagram

The global use case diagram is a representation of the structure of the main functionalities required by the users of the system.

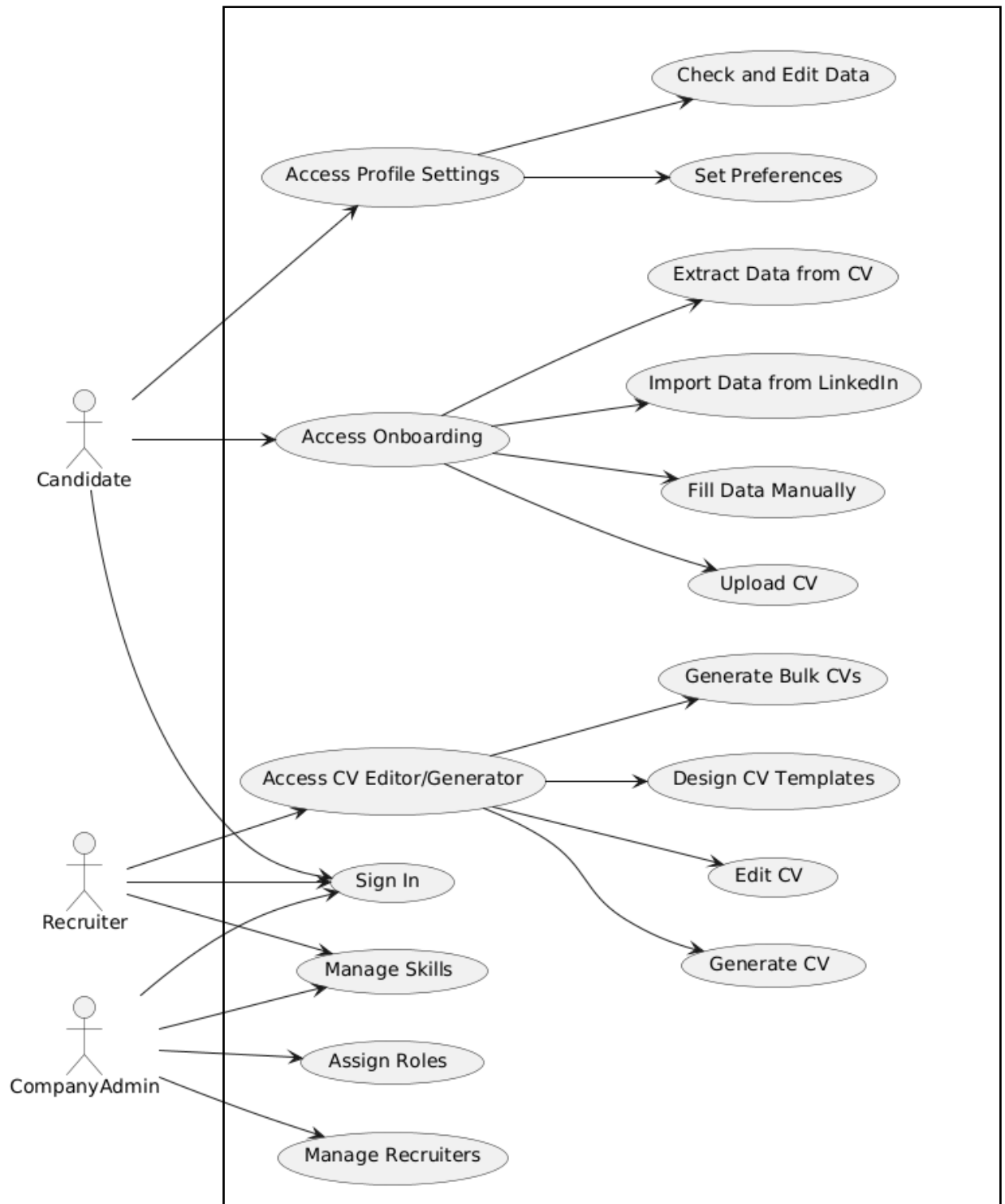


Figure 2.1 – General Use Case Diagram.

The **Figure 2.1** illustrates the general use case diagram for our project, highlighting key functionalities provided by the system. It includes the following primary use cases:

Candidate Functionalities

- **Sign In:** Grants recruiters access to system functionalities upon successful authentication.
- **Access Profile Settings:** Allows candidates to check and edit their data and set personal preferences.
- **Access Onboarding:** Includes functionalities to:
 - Import data from LinkedIn.
 - Extract data from an uploaded CV.
 - Fill in profile information manually.
 - Upload a CV.

Recruiter Functionalities

- **Sign In:** Grants recruiters access to system functionalities upon successful authentication.
- **Manage Skills:** Involves creating, updating, deleting, and searching for skills by name, as well as retrieving all questions linked to a specific skill.
- **Access CV Editor/Generator:** Enables recruiters to:
 - Edit CVs.
 - Generate CVs using pre-defined templates.
 - Design customized CV templates.
 - Generate bulk CVs.

Company Administrator Functionalities

- **Sign In:** Grants administrators access upon authentication.
- **Manage Skills:** Includes skill-related operations similar to recruiter functionalities.
- **Manage Recruiters:** Allows for the management of recruiters, including assigning roles and responsibilities within the system.

General Notes

- All functionalities require prior authentication and authorization.
- The **Company Administrator** oversees all recruiters in the organization and ensures smooth operational management of the system.

2.4 Product Backlog and Sprints Planning

The design phase is a critical phase in the life of the system. Since we are using Scrum as our methodology to develop this project, we will divide our project into sprints. We will provide the backlog and design for each sprint.

2.4.1 Product Backlog

The Product Backlog is the most important piece of Scrum artefact. The table 2.1 shows the product backlog for our project, with the various tasks to be performed.

Table 2.1 – Product Backlog.

ID	Sub-module	User stories	Priority
1	Design	As a developer, I have to have to adopt the application to its architecture and environment.	1
2	Management of project	As a developer, I have to create a branch in GitHub for upload the code and manage the versions.	1
3	Profile Management	As a Candidate, I want to view my profile settings to check my data.	2
		As a Candidate, I want to update my profile settings to edit or customize my data.	2
		As a Candidate, I want to set preferences (e.g., preferred job location) to personalize my experience.	2
4	Onboarding	As a Candidate, I want to upload my CV in PDF format for processing.	3
		As a Candidate, I want to import my data directly from LinkedIn to save time.	3
		As a Candidate, I want the extracted data to be auto filled in its correspondent field.	3
		As a Candidate, I want to manually check my details and correct them in case of mistakes.	3
5	CV Editor/Generator	As a Recruiter, I want to design a CV template that matches my preferences.	4
		As a Recruiter, I want to use or edit an existing CV template to customize the content or format.	4

		As a Recruiter, I want to generate bulk CVs for candidates in a specific job offer.	4
		As a Recruiter, I can delete my templates if I need to	4
		As a Company I want to manage my team's templates to optimize the team's dashboard	5

2.4.2 Sprints Planning

Following the presentation of the product backlog, all the sprints were identified. The release of this project was divided into two main releases, with each release have multiple sprints. The table 2.2 provides an overview of the planning for each release.

Table 2.2 – Releases Planning.

	Content of the sprint
Release 1	LinkedIn data extraction and retrieval.
	An onboarding form powered by AI-driven data extraction.
	A profile management page with all the necessary settings.
Release 2	A highly interactive template library page.
	A feature-rich and user-friendly PDF designer.
	Custom plugins for the designer, tailored specifically for CV creation.

2.5 Architectural Design

Before starting to build our project, it is important to specify the architecture of the platform, both logical and physical.

2.5.1 Physical Architecture

Starting with the physical architecture, the physical architecture used in the creation of this solution is 3-tiers architecture.

Figure 2.2 represents the 3-tiers architecture.

- **Client** : This is the web browser that allows the user to access the various man-machine interfaces of our application.

- **Application Server** : (also called middleware) the server responsible for providing the resource, but calling on another server.
- **Database Server** : MySQL is a relational and object database management system that stores the data and provides the application with the data it needs.

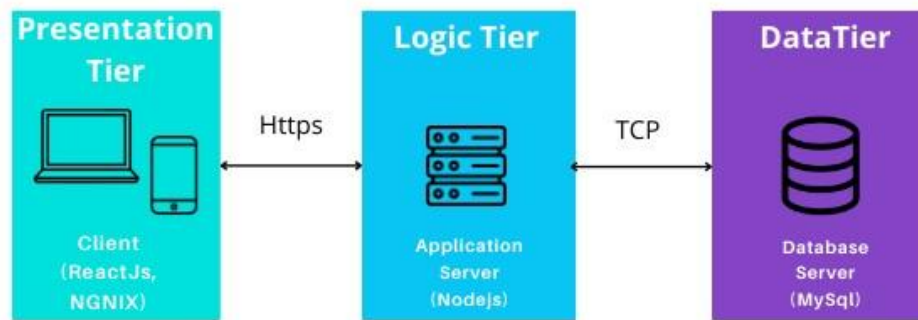


Figure 2.2 – 3 Tiers Architecture.

2.5.2 Logical Architecture

To better understand our project, we will present the overall logical architecture adopted. Figure 2.3 illustrates the interaction between the different components. we have used express JS which is a Node.js web framework that uses an MVC (model-view-controller) model.

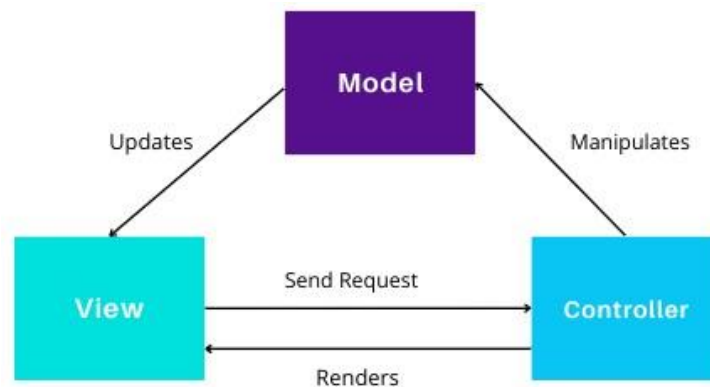


Figure 2.3 – MVC Architecture.

- **Models** : Models are objects that represent the data of the application. They are usually stored in JSON files or in a database.
- **Controllers** : Controllers are functions that handle user requests. They use models to access data and views to display data.
- **Routes** : Routes are mappings between URLs and controllers. They allow the application to know which controller to call to process a particular request.

2.6 Technological choices and requirements

The choice of technologies is an important stage in the development of an application.

2.6.1 Hardware Environment

This project has been developed using a laptop that has the following features :

- Machine : LENOVO Laptop.
- Processor : Intel(R) Core(TM) i7-10750H CPU @ 2.60GHz 2.59 GHz
- RAM : 16 GB.
- Hard Disk : 1.25 TB.

2.6.2 Software Environment

In order to create this project, we have used the following technologies in :

— **Presentation layer :**

- **React.js** : is an open-source JavaScript library for building user interfaces (UIs) and user interface components. It was originally developed by Facebook [20].
- **React Bootstrap** : is an open source library that combine React.js and Bootstarp which is a CSS framework [10].

— **Application layer :**

- **Node.js** : is an open-source, server-side runtime environment that allows developers to build and run JavaScript applications outside the web browser [21].
- **Express.js** : is open-source web application framework for Node.js. It provides a minimal and flexible set of tools for building web and mobile applications, APIs, and other networked software [3].
- **Sequelize** : Sequelize is a modern TypeScript and Node.js ObjectRelational Mapping (ORM) that supports a wide range of RDBMS such as Oracle, Postgres, MySQL ... [12].
- **Swagger** : Swagger is an open source framework that produces a suite of powerful yet easy-to-use API development tools. It generates the APIs, design and documentation needed to successfully test and deploy the application throughout its life cycle [13].

— **Data layer :**

- **MySQL :** MySQL is a popular open source relational database management system (RDBMS). MySQL can be used as a standalone client or in conjunction with other services to provide database connectivity [7].

2.6.3 Development Tools

- **VS Code :** is an open source code editor [19].
- **Postman :** is a testing tool for API [9].

2.6.4 Version Control Tools

- **Git :** Git is an open source, distributed version control system [4].

2.6.5 Deployment Tools

- **Docker :** Docker is a platform designed to assist developers in the management of container applications [2].

2.6.6 Project Management Tools

To manage the communication and the progress of the project, we use :

- **Trello :** Trello is a collaborative online project management tool. It is based on the organisation of projects into boards, which list cards, each of which represents a task [18].
- **Microsoft Teams :** Microsoft Teams is a SaaS-based application for collaborative communication that was created by Microsoft [6].

2.7 Conclusion

This chapter was devoted to identifying the functional and non-functional requirements and the actors in our system. We also listed the functionalities in the product backlog and specified our work schedule. We have also presented the architecture and the technologies to be used to develop this project.

CHAPTER 3

Implementation Of The First Release

3.1 Introduction

In this chapter we will present the first release with all its sprints. In each sprint we will present the modeling part and the implementation part with some screenshots of the interfaces.

3.2 First Sprint : "LinkedIn Data Extraction and Retrieval"

In this sprint, we focus on extracting and retrieving data from LinkedIn profiles. This allows candidates to import their professional data, such as work experience, skills, and education, directly into the platform.

3.2.1 Backlog of First Sprint

1. As a Candidate, I want to import my data directly from LinkedIn to save time.

- **Tasks:**
 - Integrate LinkedIn API to allow candidates to link their LinkedIn profiles.
 - Ensure that the LinkedIn data, such as work experience, education, and skills, can be retrieved and mapped correctly to the platform's profile fields.
 - Develop the UI elements (e.g., "Import from LinkedIn" button) to initiate the data import.
 - **Test and debug** the functionality to ensure smooth operation and user experience.

2. As a Candidate, I want the extracted data to be auto-filled in its corresponding fields.

- **Tasks:**
 - Implement the logic for extracting data from LinkedIn and mapping it to the corresponding profile fields (e.g., job titles, work experience, skills, education).
 - **Test and debug** the mapping process to ensure correct data extraction and placement into fields.
 - Ensure that the data retrieved is pre-populated into the candidate's profile, reducing manual entry.
 - **Test and debug** pre-population of fields to ensure accuracy and completeness.

3. As a Candidate, I want to manually check my details and correct them in case of mistakes.

- **Tasks:**
 - Develop an interface that displays imported data from LinkedIn.
 - **Test and debug** the interface to ensure that it displays the data correctly.

- Provide options to edit or correct any automatically populated fields in case of discrepancies or incomplete data.
- **Test and debug** the manual editing functionality to ensure smooth user experience.

Deliverables:

- LinkedIn data import functionality.
- A system for mapping LinkedIn data to profile fields.
- A candidate profile view where imported data is displayed with editable fields.

3.2.2 Analysis of the First Sprint

In this section, we focused on the analysis part of the first sprint. **Figure 3.1** represents the use Sequence diagram of the first sprint. It can be observed that the candidate is able to import their LinkedIn data into the platform. This includes connecting their LinkedIn profile, extracting relevant information such as work experience, skills, and education, and auto-filling their profile fields.

Additionally, the candidate can manually review and correct any auto-filled data to ensure accuracy. This feature provides flexibility in handling discrepancies or incomplete details. Furthermore, recruiters can view the imported candidate profiles with the updated LinkedIn data for recruitment purposes.

The platform ensures robust functionality by providing a seamless data mapping process and a user-friendly interface for manual adjustments.

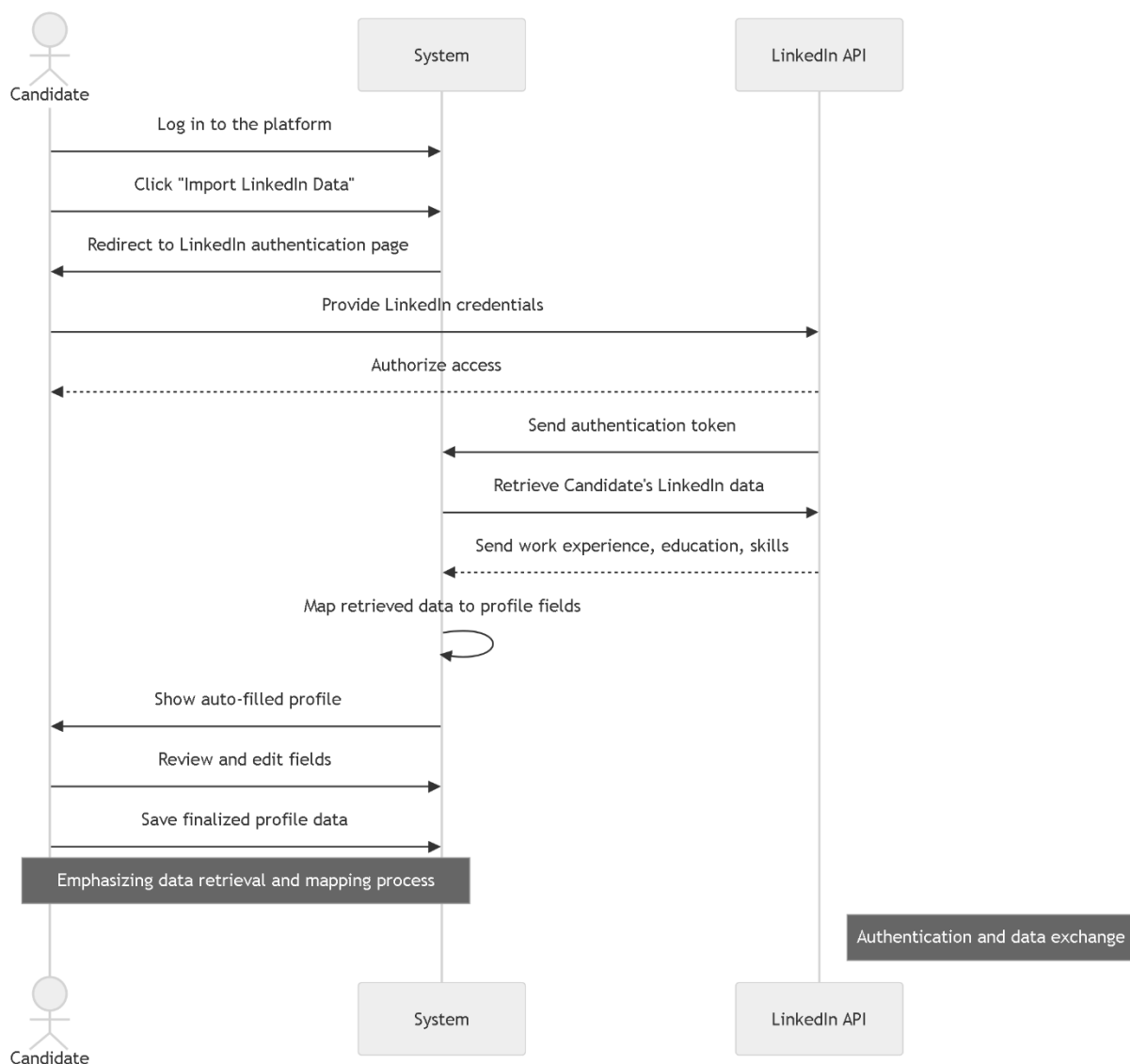


Figure 3.1 – Sequence Diagram of " LinkedIn data extraction and retrieval".

Table 3.1 – Description of the " LinkedIn data extraction and retrieval " use case.

Use case	LinkedIn data extraction and retrieval
Actor	Candidate
Pre-condition	The candidate is logged into the system and has a valid LinkedIn account.

Nominal scenario	1. The candidate is redirected to the onboarding section once his account is verified. 2. The candidate selects the "Import LinkedIn Data" option. 3. The system redirects the candidate to log in to their LinkedIn account via OAuth. 4. After successful authentication, the system retrieves basic data including full name, profile picture and most important LinkedIn account username and ID. 5. With candidate consent granted, we use the data provided by the API to access the user LinkedIn profile and scrape it to get data such as work experience, education, skills ...etc. 6. Extracted data is mapped to corresponding fields in the system. 7. The candidate reviews the imported data and adjusts it as needed. 8. The candidate can: - submits the finalized form, saving the data to their profile. - Skip the process expecting to be back on his next login. - Note it for later so he can manually get back later but never be redirected back on login.
Exceptional scenario	If an error occurs during import (for example, data is missing or incorrectly formatted file ...),
	The application displays an error message and allows the user to correct the problems before moving on.

3.2.4 Implementation of The First Sprint

In this section, we will present the implementation of the first sprint.

LinkedIn data retrieval

As shown in the Figure 3.2 you can import your personal and professional information directly from your LinkedIn making the onboarding process a lot easier for the new users.

Help us know you more

Logout |  

General information

First Name Last Name

Country Phone Number Language

Email LinkedIn GitHub

Area of Expertise

Technical Skills

Soft Skills

Experiences

[Note for later](#) [Skip>>](#)

Figure 3.2 – Onboarding Page’s LinkedIn data import button.

Once you click the button, you will be redirected to the LinkedIn authentication page to provide consent for us to access the necessary data from your LinkedIn profile.

LinkedIn

HireCue would like to:

- Use your name and photo
- Use the primary email address associated with your LinkedIn account

You can stop this sync in your LinkedIn [settings](#). Matchory [terms](#) apply. [Learn more](#).

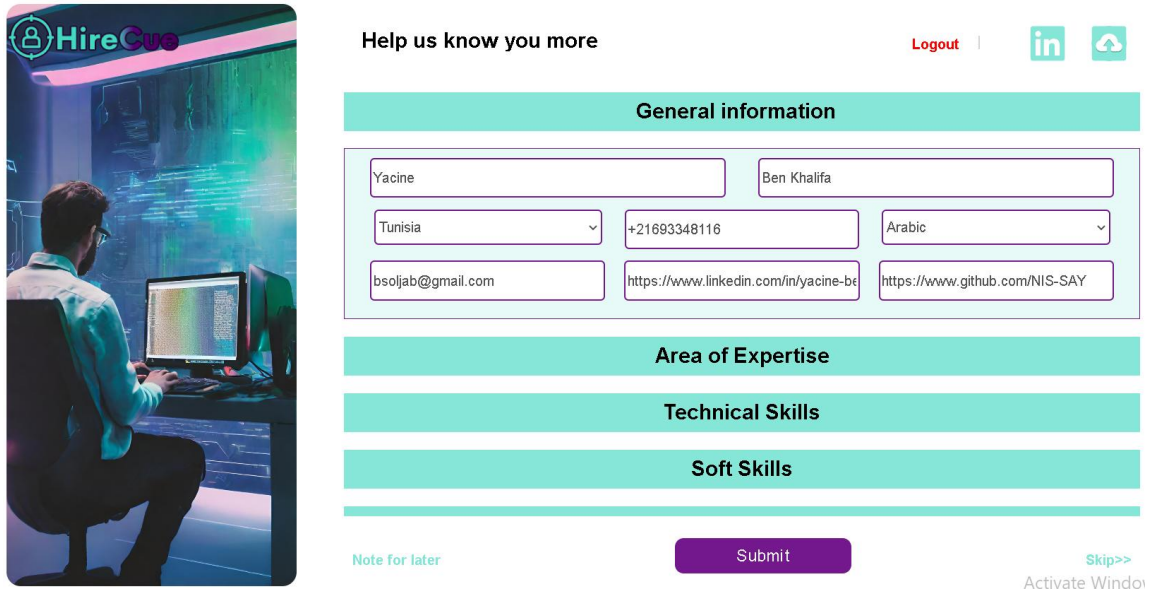
[Not you?](#)

You will be redirected to <https://app.hirecue.com>

[Privacy Policy](#) [User Agreement](#)

Figure 3.3 – LinkedIn authentication.

After completing the authentication process, you will be redirected to the onboarding page, where the retrieved data will be automatically populated into the corresponding fields..



The screenshot shows the HireCue onboarding interface. On the left, a person is working at a computer. The main form is titled 'Help us know you more' and includes a 'Logout' link and social media icons. The form is divided into sections: 'General information', 'Area of Expertise', 'Technical Skills', and 'Soft Skills'. The 'General information' section contains pre-filled fields for Name (Yacine Ben Khalifa), Location (Tunisia), Phone Number (+21693348116), Language (Arabic), Email (bsoljab@gmail.com), LinkedIn profile (https://www.linkedin.com/in/yacine-ben-khalifa), and GitHub profile (https://www.github.com/NIS-SAY). At the bottom, there is a 'Submit' button and a 'Skip>>' link with the text 'Note for later' and 'Activate Window'.

Figure 3.4 – LinkedIn data is auto filled in the correspondent fields.

3.3 Second Sprint : "An Onboarding Form Powered by AI-Driven Data Extraction"

In the second sprint, we focus on improving the onboarding process by using AI-driven data extraction from CVs uploaded by candidates. This feature will help automatically extract and fill in the candidate's details, such as contact information, skills, and work experience, using AI.

3.3.1 Backlog of Second Sprint

1. As a Candidate, I want to upload my CV in PDF format for processing.

- **Tasks:**

- Develop an upload functionality for candidates to submit their CVs in PDF format.
- **Test and debug** the CV upload functionality to ensure smooth file submission and validation.
- Implement backend logic to store and process the uploaded files.
- **Test and debug** backend processing to ensure the correct handling of PDF files.

2. As a Candidate, I want the extracted data to be auto-filled in its corresponding fields.

- **Tasks:**
 - Integrate AI-powered data extraction tools (e.g., using Natural Language Processing) to parse and extract relevant details from uploaded CVs.
 - **Test and debug** the AI extraction process to ensure accuracy in the extracted data.
 - Develop functionality to automatically fill in the extracted information into the profile fields (such as name, contact info, work experience, and skills).
 - **Test and debug** the auto-fill feature to ensure the data is correctly placed in each field.

3. As a Candidate, I want to manually check my details and correct them in case of mistakes.

- **Tasks:**
 - Provide a preview of the extracted data and give candidates the option to review and edit the details.
 - **Test and debug** the preview functionality to ensure the data is displayed correctly.
 - Implement functionality for candidates to make changes to any field that was auto-filled incorrectly or incompletely.
 - **Test and debug** the editing functionality to ensure smooth user interaction.

Deliverables:

- CV upload functionality with PDF support.
- AI-driven data extraction that populates profile fields.
- A UI that allows candidates to review and edit extracted data.

3.3.2 Analysis of the Second Sprint

In this section, we will focus on the analysis part of the second sprint. Figure 3.5 represents the use case diagram of the second sprint. It highlights that the onboarding process is powered by AI-driven data extraction. Candidates can upload their CVs in PDF format or import their data directly from LinkedIn. The system processes the uploaded or imported data using AI to extract relevant details, such as work experience, education, and skills. Furthermore, the system auto-fills the onboarding form fields with the extracted data. Candidates are provided with an interface to manually review and correct any inaccuracies or missing information before finalizing their profile. This sprint streamlines the onboarding process by reducing manual effort and enhancing accuracy through AI integration. Table 3.3 represents a detailed description of the "Upload CV and process data" use case.

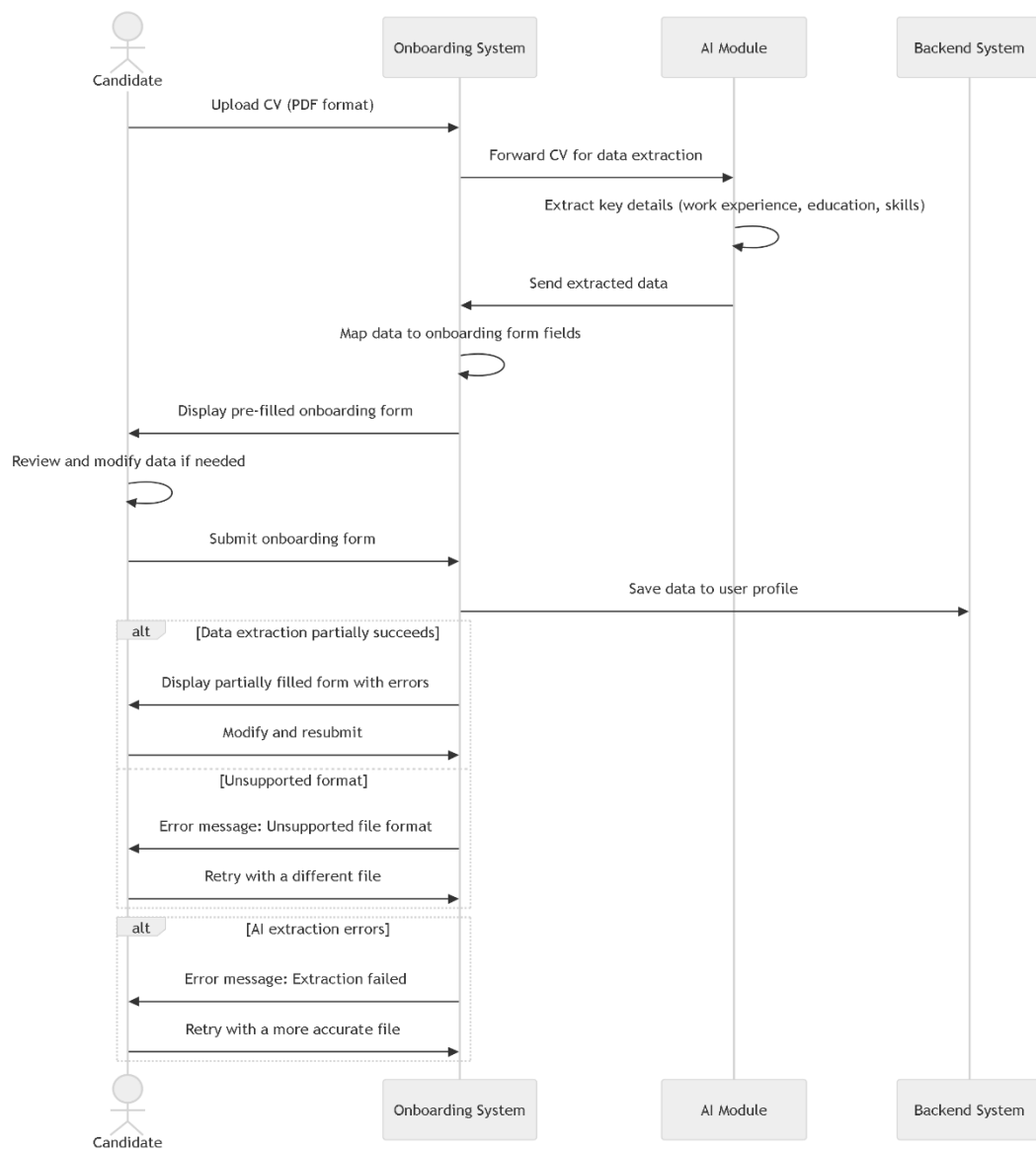


Figure 3.5 – Sequence Diagram of " Upload CV and process data ".

Table 3.2 – Description of the " Upload CV and process data " use case.

Use case	Upload CV and process data
Actor	Candidate
Pre-condition	The candidate is logged into the system.

Nominal scenario	1. The candidate is redirected to the onboarding section once his account is verified. 2. The candidate uploads a CV file in PDF format. 3. The system processes the uploaded CV or LinkedIn data using AI to extract relevant information. 4. Extracted data, including work experience, education, and skills, is auto-filled into the corresponding fields in the onboarding form. 5. The candidate reviews the auto-filled data and makes manual adjustments if necessary. 6. The candidate can: - submits the finalized form, saving the data to their profile. - Skip the process expecting to be back on his next login. - Note it for later so he can manually get back later but never be redirected back on login.
Exceptional scenario	If an error occurs during import (for example, data is missing or incorrectly formatted file ...),
	the application displays an error message and allows the user to correct the problems before moving on.

3.3.3 Implementation of The Second Sprint

In this section, we will present the implementation of the second sprint.

Onboarding Page

Figures 3.7 and 3.8 present the dedicated sections where users can manually fill in their personal and professional information.

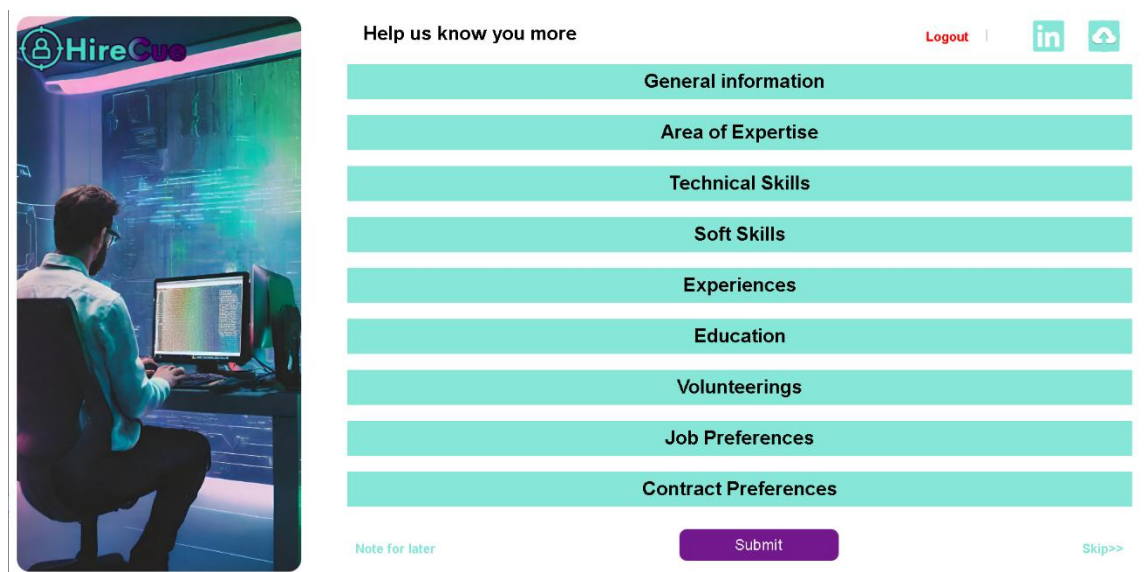


Figure 3.7 – Onboarding Page’s different sections.

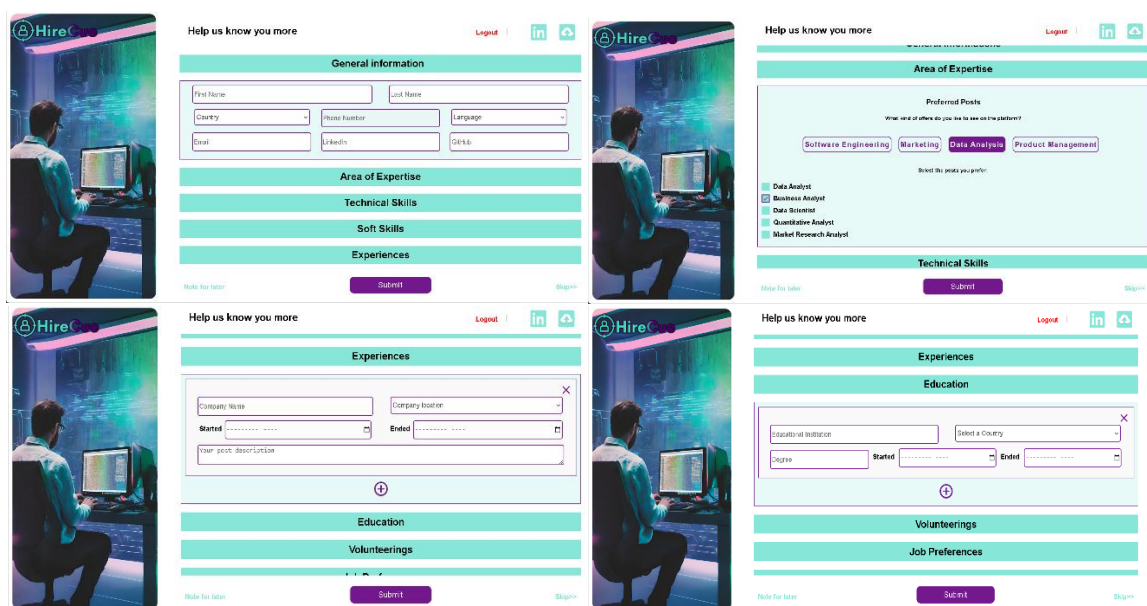


Figure 3.8 – Onboarding Page’s sections overview.

If a user attempts an unauthorized action, proceeds without providing the required information, or encounters an error, pop-ups with instructions or error messages will appear to guide them in the right direction.

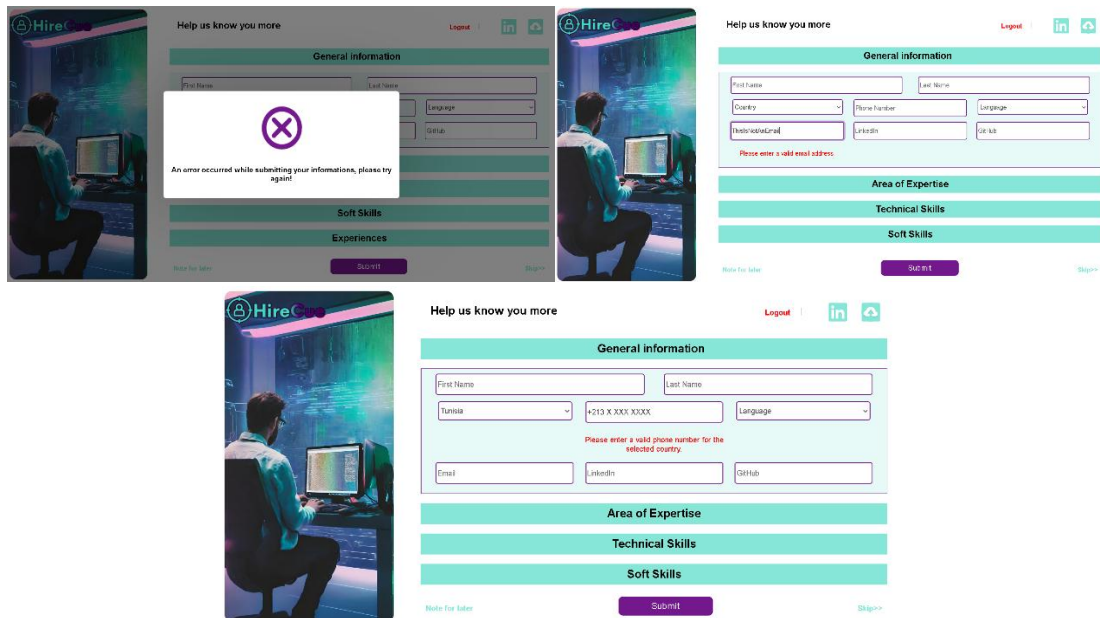


Figure 3.9 – Onboarding Page's error handling.

Users can click the 'Upload CV' button to upload their CV for data extraction.

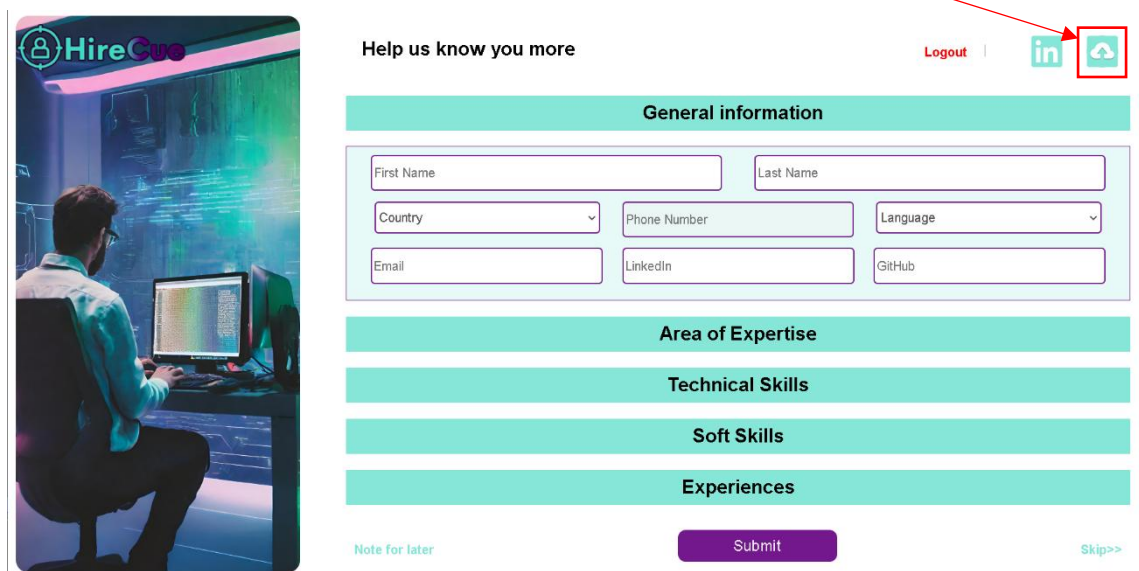


Figure 3.10 – Onboarding Page's CV data extraction button.

Then, a pop-up will appear where the user can drag and drop, copy and paste, or browse for their CV PDF file, and confirm to start the data extraction process.

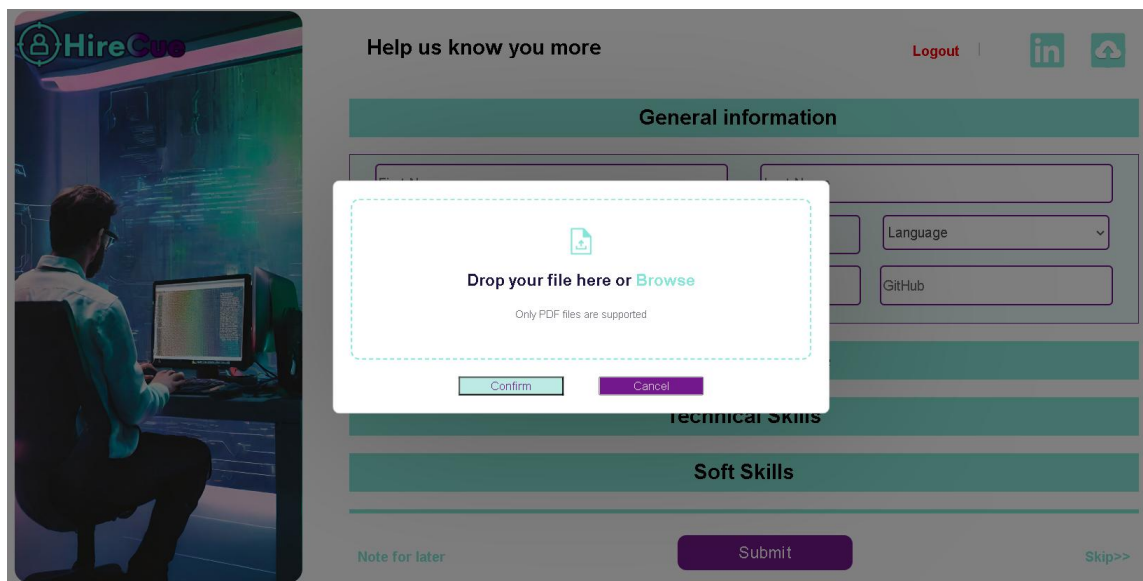


Figure 3.11 – Onboarding Page’s CV upload interface.

Once confirmed, a loader will be displayed on the screen while the extraction results are being retrieved.

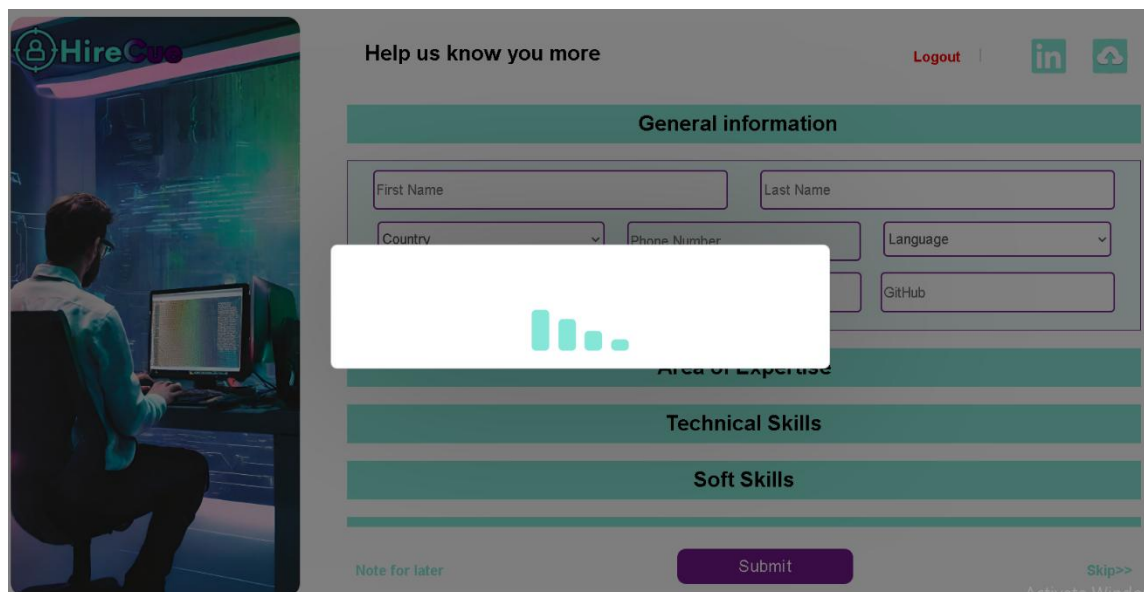


Figure 3.12 – Data extraction loader.

When the results are received from the AI server, a success pop-up will replace the loader.

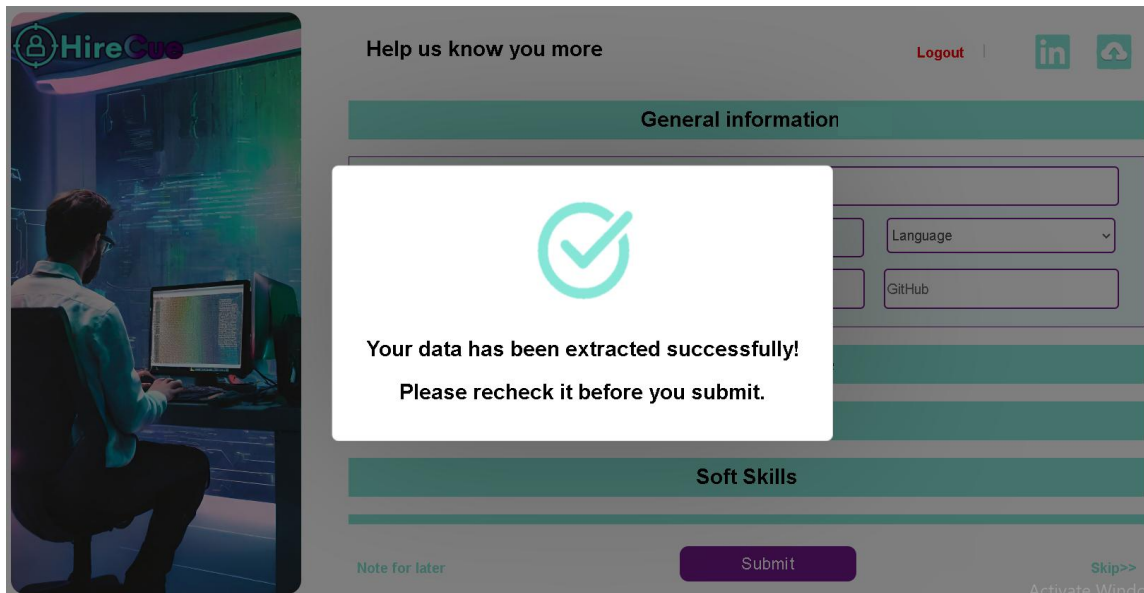


Figure 3.13 – Data extraction confirmation.

The user can now click anywhere outside the pop-up to close it and find the retrieved information mapped across the page sections.

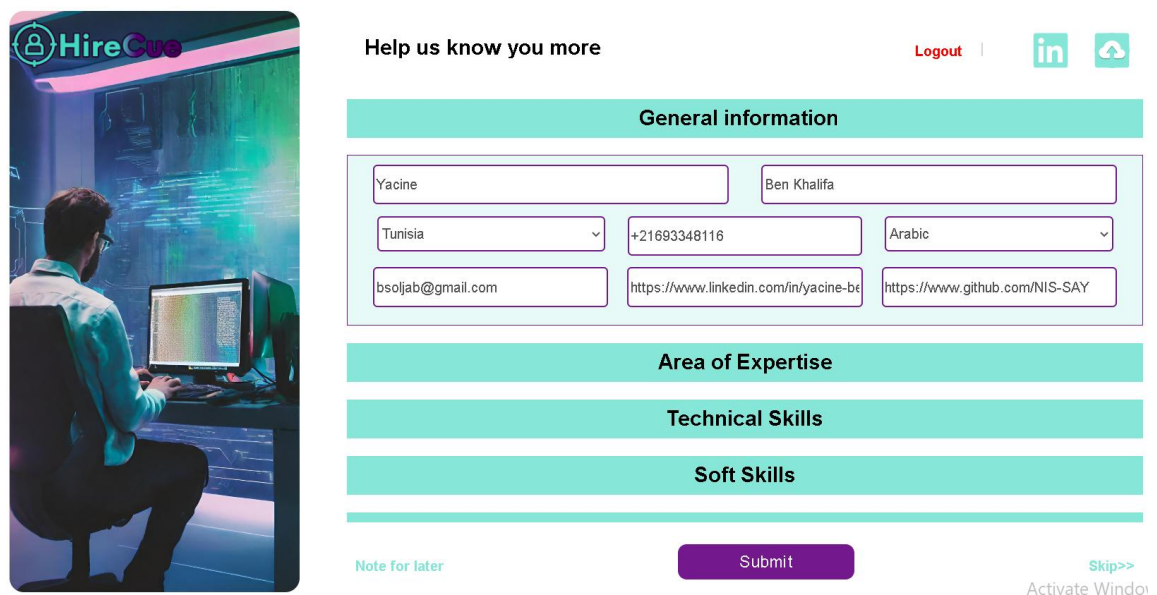


Figure 3.14 – Extrcted data is auto filled in the correspondent fields.

3.4 Third Sprint : "A Profile Management Page with All the Necessary Settings"

In the third sprint, we will implement the Profile Management page, where candidates can view, update, and manage their profile settings, preferences, and other personal information.

3.4.1 Backlog of Second Sprint

1. As a Candidate, I want to view my profile settings to check my data.

- **Tasks:**
 - Develop a "Profile Management" page where candidates can view their details, including work experience, skills, and education.
 - **Test and debug** the profile display functionality to ensure all fields are correctly populated and displayed.
 - Display the extracted and manually entered data in an easily readable format.
 - **Test and debug** the display layout and ensure it is user-friendly.

2. As a Candidate, I want to update my profile settings to edit or customize my data.

- **Tasks:**
 - Create an "Edit" functionality that allows candidates to update their profile information (such as contact details, work experience, skills, etc.).
 - **Test and debug** the editing functionality to ensure changes are saved correctly and the data is updated in the system.
 - Ensure all changes made by the candidates are saved and updated in the system.
 - **Test and debug** the saving and updating process to ensure accuracy.

3. As a Candidate, I want to set preferences (e.g., preferred job location) to personalize my experience.

- **Tasks:**
 - Develop a section in the profile settings where candidates can specify preferences (e.g., preferred job location, salary expectations, job types).
 - **Test and debug** the preferences section to ensure that candidates can save and apply their preferences without issues.
 - Implement functionality to save and apply these preferences during job search or matching.

- **Test and debug** the preference-saving and application functionality to ensure that they are correctly applied in the platform.

Deliverables:

- A Profile Management page that displays the candidate's profile and allows editing.
- A preferences section for candidates to personalize their experience.
- A system for saving and updating profile data.

3.4.2 Analysis of the Third Sprint

In this section, we analyze the third sprint, focusing on the Profile Management feature. The goal of this sprint was to enable candidates to manage their profile settings effectively. This functionality allows candidates to view, update, and save their personal information on the platform, ensuring a seamless and secure user experience.

The Profile Management page provides an intuitive interface for candidates to modify details such as their name, contact information, preferred job location, and other profile-related settings. Figure 3.15 illustrates the Sequence Diagram for this sprint.

Table 3.3 represents a detailed description of the "Profile Management Page with All the Necessary Settings" use case.

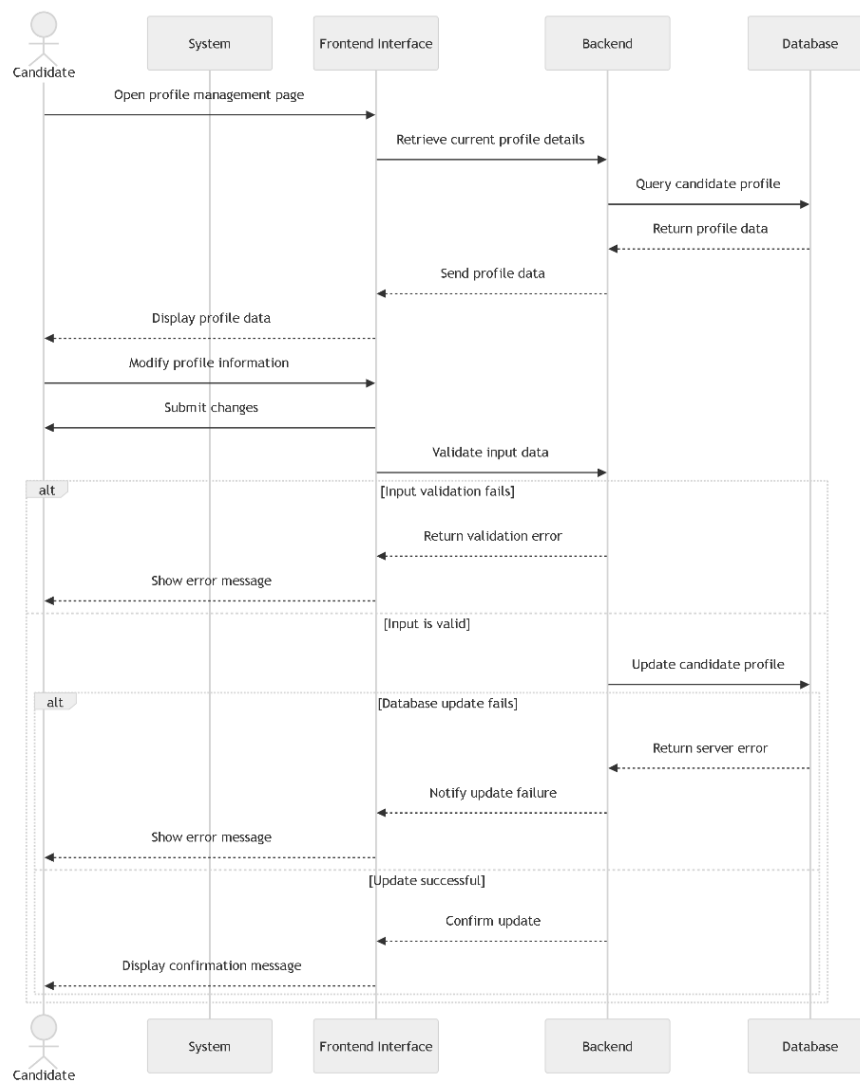


Figure 3.15 – Sequence Diagram of " A Profile Management Page with All the Necessary Settings ".

Table 3.3 – Description of the " A Profile Management Page with All the Necessary Settings " use case.

Use case	A Profile Management Page with All the Necessary Settings
Actor	Candidate
Pre-condition	The candidate is logged into the system.

Nominal scenario	1. The candidate navigates to the "Profile Management" page. 2. The system retrieves the candidate's current profile details from the database. 3. The profile management interface displays the candidate's details (e.g., name, email, phone number, preferred job location). 4. The candidate makes modifications to their profile details, such as updating their phone number or setting a preferred job location. 5. The candidate clicks the "Save" button. 6. The system validates the input data (e.g., correct format for phone numbers, security breach attempts like SQL injection or else). 7. The backend updates the candidate's profile information in the database.
Exceptional scenario	If an error occurs during import (for example, data is missing or incorrectly formatted file ...),
	the application displays an error message and allows the user to correct the problems before moving on.

3.4.3 Implementation of The Third Sprint

In this section, we will present the implementation of the second sprint.

Profile Page

Figure 3.16 and Figure 3.17 presents the interfaces in which we can view and edit your personal and professional information in the dedicated sections to make sure our profile is up to date causing more accurate job matching .

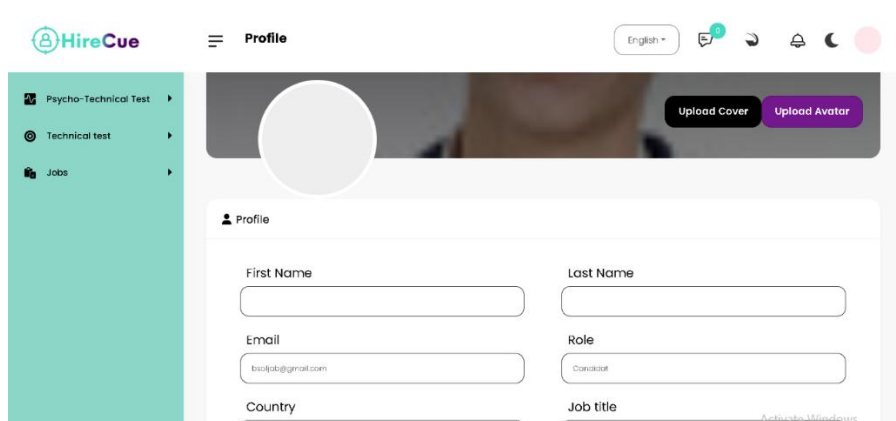


Figure 3.16 Profile management page.

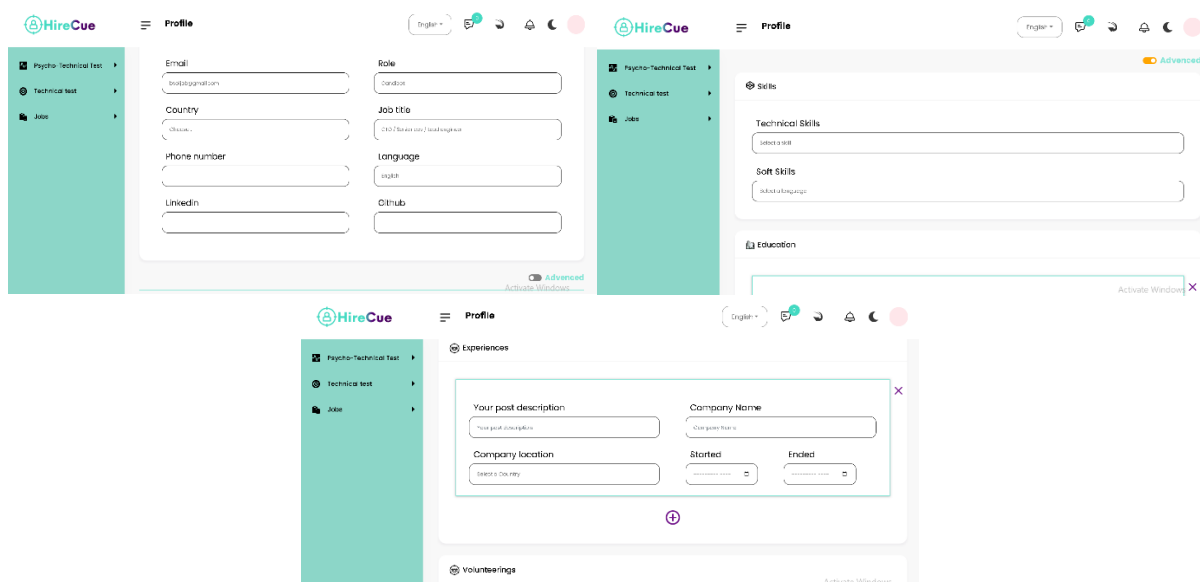


Figure 3.17 Different profile sections.

The last section is reserved for the user's preferences, where they can further personalize their matchmaking.

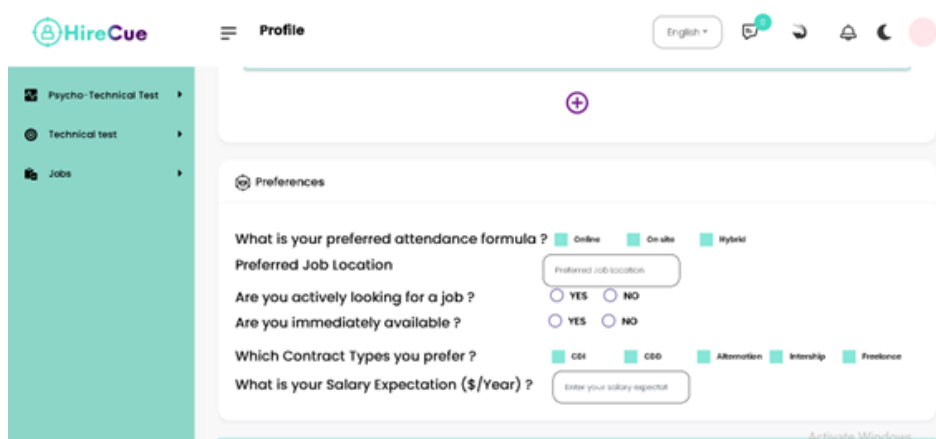


Figure 3.18 Preferences management section.

3.5 Conclusion

Using screenshots of the interfaces, this chapter presented the implementation of the first release with all its sprints and their modeling.

CHAPTER 4

Implementation Of The Second Release

4.1 Introduction

In this chapter, we will introduce the second release along with all its sprints. For each sprint, we will cover the modeling and implementation phases, accompanied by screenshots of the interfaces.

4.2 Fourth Sprint : " A highly interactive template library page"

In this sprint, we focus on building a highly interactive and user-friendly template library page. This allows recruiters and company admins to access, browse, and manage templates with ease. The page will feature three main sections: "My Templates," "My Company's Templates," and "Suggestions" created by platform developers. Users will be able to clone existing templates for further customization, ensuring a seamless and efficient experience while maintaining strict permissions based on their role. This sprint aims to enhance the template management system, providing a smooth and intuitive interface for users to start their CV creation process.

4.2.1 Backlog of First Sprint

1. As a Recruiter, I want to view my own templates in a dedicated section.

- **Tasks:**
 - Develop the "My Templates" section to display templates uploaded or saved by the recruiter.
 - Fetch and display only templates associated with the recruiter.
 - Test and debug to ensure accurate fetching and categorization of templates.
 - Ensure the layout is user-friendly and templates are easily accessible.

2. As a Company Admin, I want to view templates created by myself and recruiters within my company.

- **Tasks:**
 - Develop the "My Company's Templates" section to display company-wide templates.
 - Fetch templates created by company admins and recruiters within the company.
 - Test and debug to confirm proper data fetching and categorization.
 - Ensure clarity in template ownership and easy navigation.

3. As a Recruiter or Company Admin, I want to explore Suggestions created by platform developers in a dedicated section.

- **Tasks:**
 - Ensure Suggestions are visually distinct and immutable.
 - Test and debug to guarantee seamless template browsing.
 - Optimize for responsive and user-friendly design.

4. As a Recruiter or Company Admin, I want to browse templates across all sections.

- **Tasks:**
 - Optimize browsing functionality with infinite scroll.
 - Test and debug the browsing experience to ensure usability.
 - Ensure smooth navigation between "My Templates," "My Company's Templates," and "Suggestions".

5. As a Recruiter or Company Admin, I want to use an existing template as a starting point for a new one.

- **Tasks:**
 - Implement a "Use Template" button to clone templates.
 - Restrict cloning to templates the user has permission to view.
 - Ensure cloned templates are saved under "My Templates" for further customization.
 - Test and debug to confirm proper functionality and template ownership

6. As a Recruiter, I want to modify or delete only my own templates.

- **Tasks:**
 - Implement permission checks to restrict editing/deleting to the recruiter's own templates.
 - Add validation for user ownership before performing these actions.
 - Test and debug to ensure permissions are enforced correctly.

7. As a Company Admin, I want to modify or delete my own templates and those created by recruiters in my company.

- **Tasks:**
 - Extend permission checks to allow admins to modify/delete templates in "My Templates" and "My Company's Templates".
 - Restrict admins from editing Suggestions or templates from other companies.
 - Test and debug role-based permission checks.

8. As a Recruiter or Company Admin, I want to view templates in all sections but cannot modify or delete templates I do not own or have permission for.

- **Tasks:**
 - Implement permission ladder logic to enforce modification restrictions.
 - Test and debug to ensure Suggestions remain immutable.
 - Ensure all templates are viewable but permissions for modification are clearly defined.

9. As an Unauthorized User, I am restricted from accessing the template library page.

- **Tasks:**
 - Implement authentication checks to restrict access to recruiters and company admins only.
 - Redirect unauthorized users to an error page or login screen.
 - Test and debug access restrictions.

Deliverables:

- **A Template Library Page with three sections:**
 - "My Templates" for recruiter-specific templates.
 - "My Company's Templates" for company-wide templates.
 - "Suggestions" for platform-curated templates.
- **Role-Based Permission System:**
 - Recruiters can modify/delete only their own templates.
 - Company admins can modify/delete their own templates and those created by company recruiters.
 - Suggestions are immutable for all users.
- **Interactive Template Browsing:**
 - Templates displayed in a user-friendly layout without search or filter options.
 - A cloning feature for creating new templates based on existing ones.
- **Access Control:**
 - Page restricted to authenticated recruiters and company admins.

4.2.2 Analysis of the Fourth Sprint

In this section, we analyze the first sprint, focusing on the Template Library feature. The goal of this sprint was to enable recruiters and company admins to access and manage templates effectively. This functionality provides users with a structured and interactive interface, divided into three sections: "My Templates," "My Company's Templates," and "Suggestions" created by platform developers. Users can clone templates for customization while adhering to role-based permissions, ensuring data integrity and proper access control.

The Template Library page offers an intuitive interface for users to browse, view, and interact with templates, streamlining the process of starting new CV designs. Figure 4.1 illustrates the Sequence Diagram for this sprint.

Table 4.1 represents a detailed description of the "Highly Interactive Template Library Page" use case.

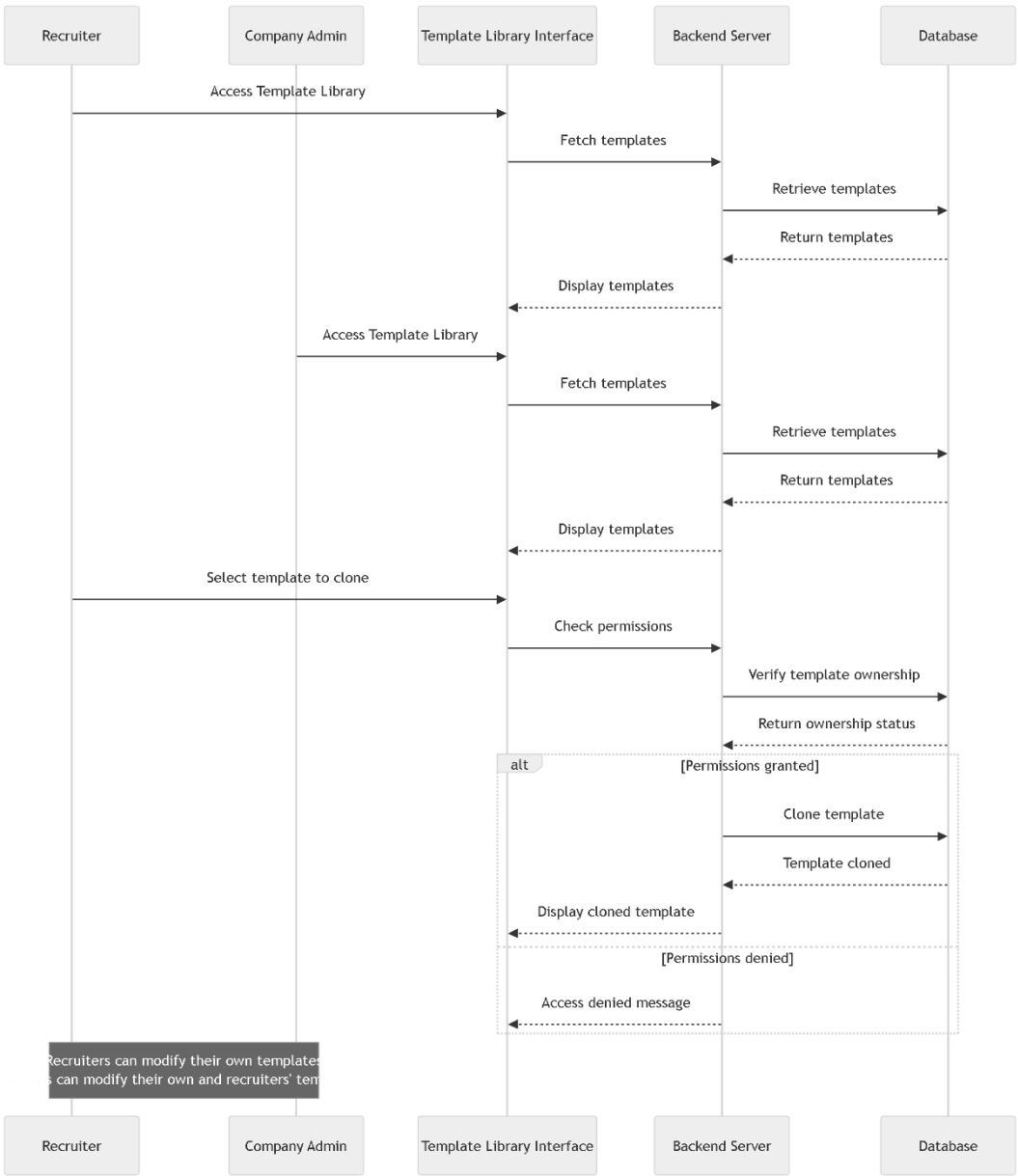


Figure 4.1 – Sequence Diagram of " Highly Interactive Template Library Page ".

Table 4.1 – Description of the " Highly Interactive Template Library Page " use case.	
Use case	A Highly Interactive Template Library Page

Actor	Recruiter, Company Admin
Pre-condition	The user (Recruiter or Company Admin) is logged into the system and has the necessary permissions to access the Template Library page.
Nominal scenario	1. The user navigates to the "Template Library" page. 2. The system automatically retrieves and displays templates categorized into three sections: ○ "My Templates" ○ "My Company's Templates" ○ "Suggestions" 3. The user selects a template from one of the sections. 4. The system checks the user's permissions: ○ Recruiters can modify their templates. ○ Company admins can modify their templates and those of their company's recruiters. ○ Suggestions cannot be modified but can be cloned into "My Templates" 5. If the user has permission, they can modify a template or use it as a starting point for creating a new one. 6. The system saves any modifications or new templates in the database. 7. Security measures (e.g., input sanitization, role-based access checks) are applied during actions to prevent SQL injection or breaches.
Exceptional scenario	If an unauthorized user tries to access the page, the system denies access and redirects them to the login or an error page
	If a user attempts an action without the required permissions (e.g., modifying a template they don't own), the system displays an error.

3.2.3 Implementation of The Fourth Sprint

In this section, we will present the implementation of the second sprint.

Template library Page

Figure 4.2 presents the interfaces in which users can find different categories of templates and use them to generate CVs or as starting point for a new template .

While waiting for the database results, the template cards are replaced with loaders.

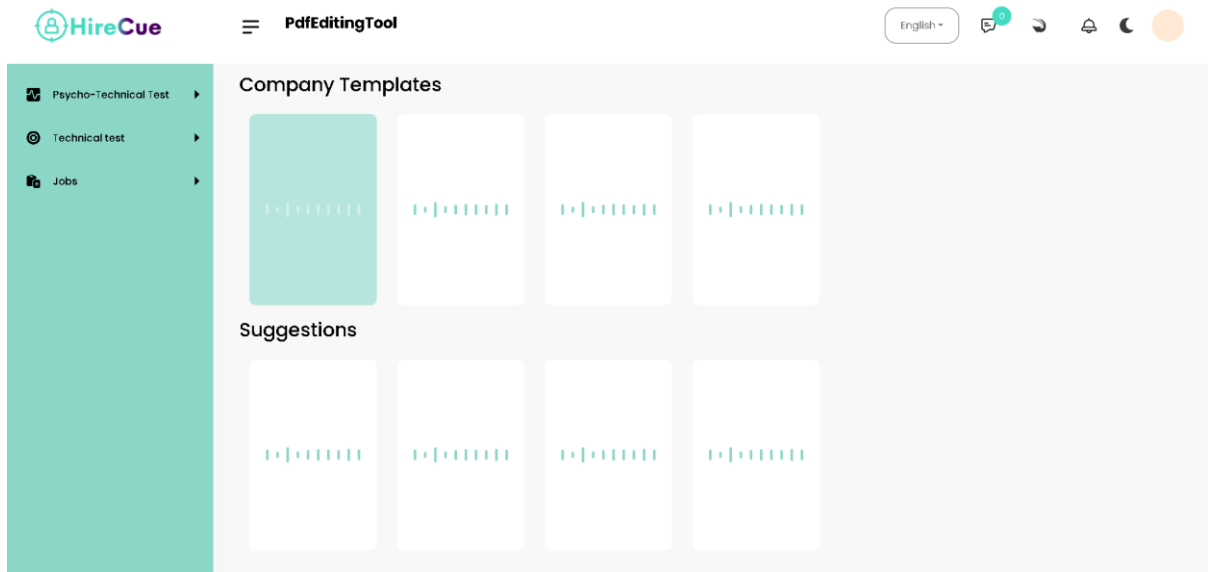


Figure 4.2 –Template Library Page while getting data.

If section templates are available, their cards will be displayed in an infinitely scrollable horizontal list.

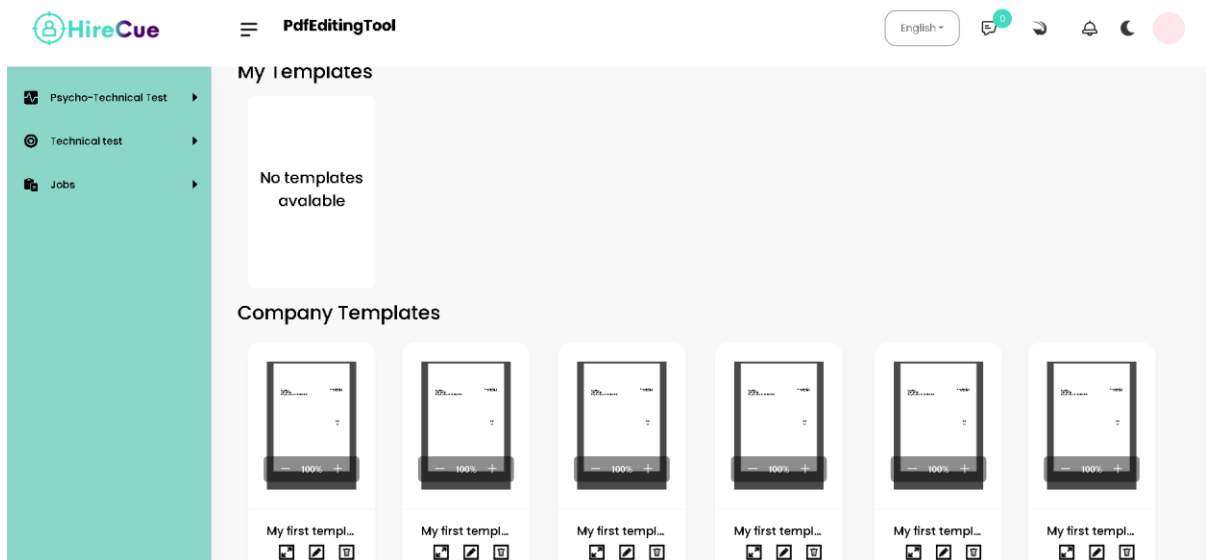


Figure 4.3 –Template Library Page.

Users can click on a template to open an options pop-up, where they can choose to use the template once, bulk generate CVs using that specific template, or edit the template to create a new one.

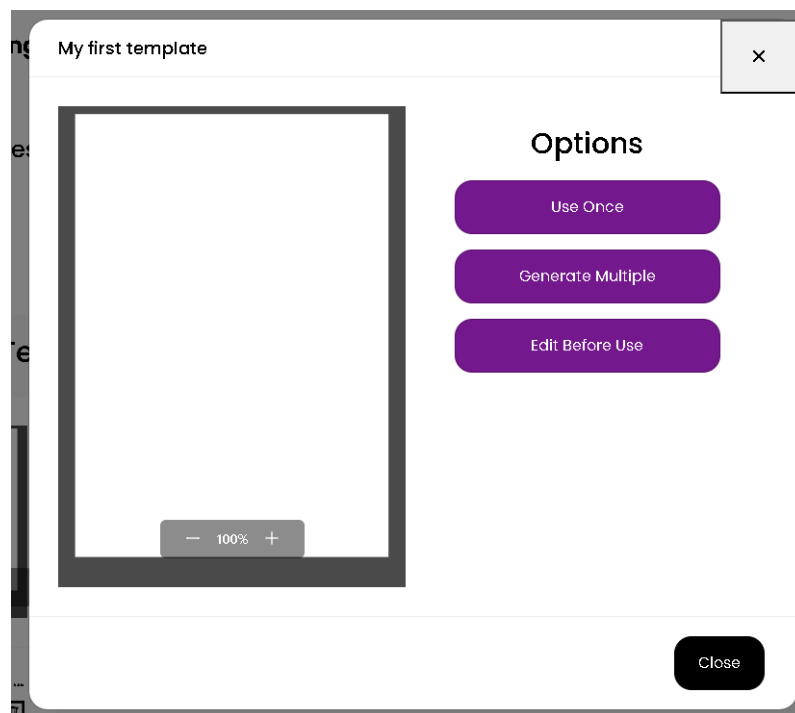


Figure 4.4 –Template options popup.

If you choose the 'Use Once' option, the pop-up switches to the 'Use Once' interface, allowing you to directly fill in the information in the preview, name the file, and then download it.

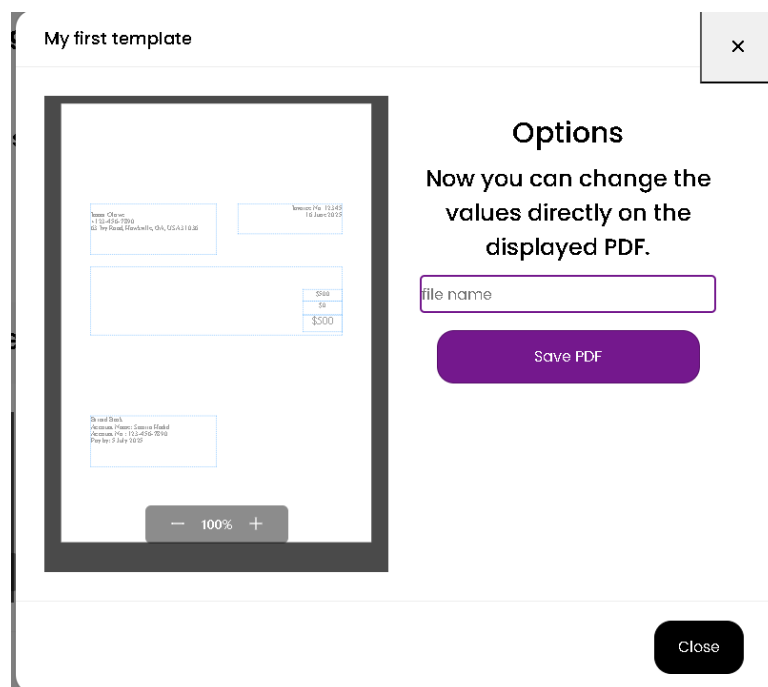


Figure 4.5 –Directly update inputs on the preview to generate one copy.

If you choose the 'Generate Multiple' option, the pop-up switches to the 'Generate Multiple' interface, allowing you to choose the data source for filling the information.

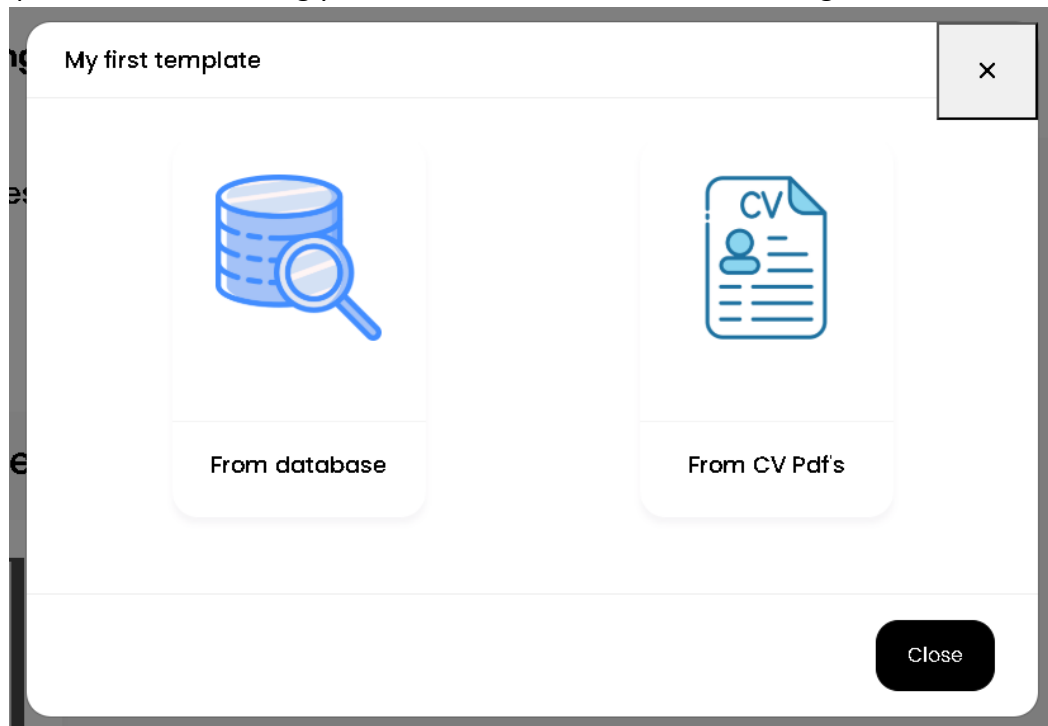


Figure 4.6 –Choose data source for bulk generation.

When you want to generate CVs for candidates from our database, all you need to do is select the job offer you want to refer to and confirm to start the generation process.

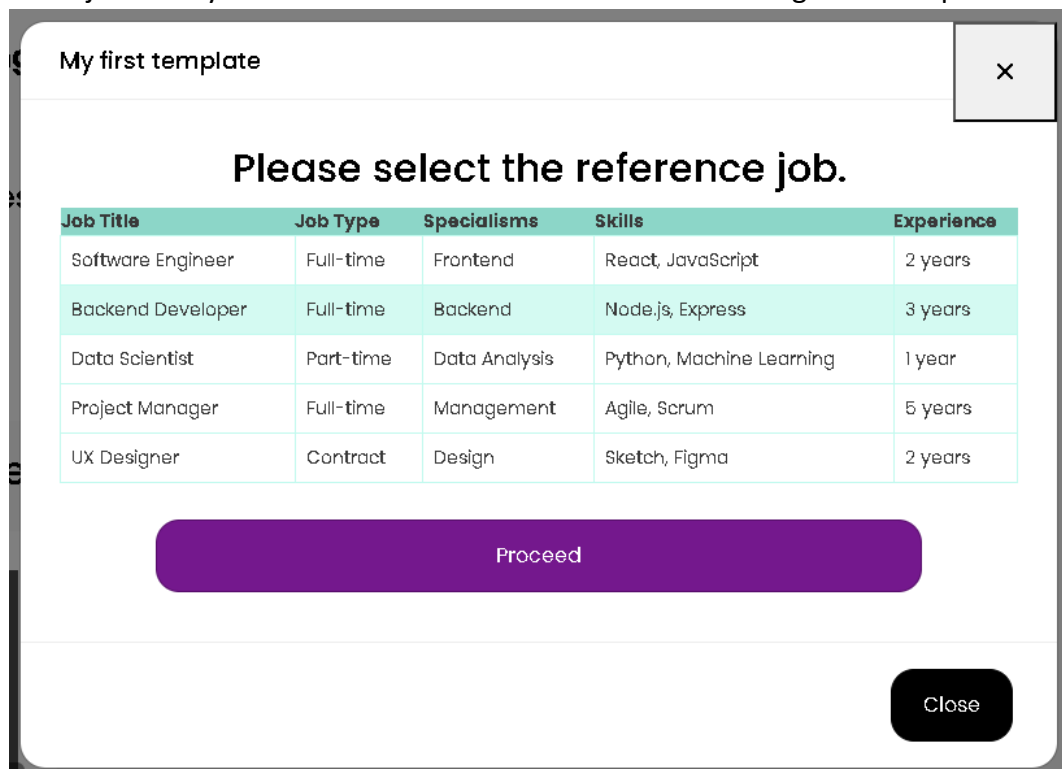


Figure 4.7 –Choose reference job offer to bulk generate from our database.

You can also generate templated CVs from locally saved CVs to standardize their format. All you need to do is provide the required files by dragging and dropping, copying and pasting, or browsing for them.

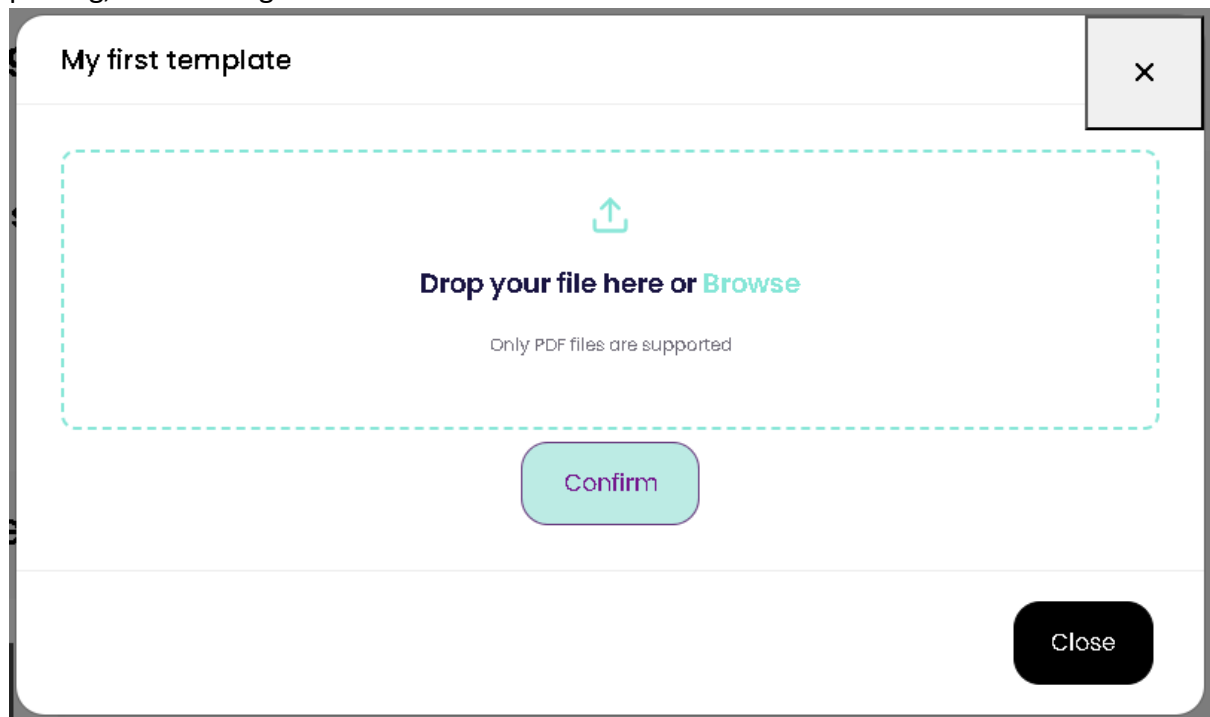


Figure 4.8 – Upload CV Pdf's to bulk generate from files.

4.3 Fifth Sprint : " A Feature-Rich and User-Friendly PDF Designer "

In this sprint, we focus on developing a highly interactive and feature-rich PDF Designer. This tool empowers recruiters and company admins to create and customize professional PDF templates with ease. The designer provides an intuitive interface, enabling users to add, modify, and style various elements such as text, images, forms, tables, and signatures. Users can also bulk-generate PDFs and customize every field, ensuring tailored documents for diverse needs.

This sprint aims to enhance user productivity and flexibility by delivering a robust and user-friendly PDF customization platform.

4.3.1 Backlog of First Sprint

1. As a Recruiter or Company Admin, I want to design a new PDF template for creating customized documents.

- **Tasks:**
 - Develop a "PDF Designer" interface with drag-and-drop functionality.
 - Implement features to add text, images, and layout adjustments on the canvas.
 - Enable real-time preview of the design.
 - Develop a backend service for saving and retrieving PDF templates.
 - Test and debug the designer's UI and ensure it is user-friendly.
 - Validate inputs to prevent XSS, SQL injection, and other security vulnerabilities.

2. As a Recruiter or Company Admin, I want to save a template for future use or editing.

- **Tasks:**
 - Implement a "Save Template" functionality to store the design in the database.
 - Ensure saved templates are accessible based on user permissions.
 - Test and debug saving functionality to ensure no data is lost.
 - Implement secure storage for uploaded assets like images.

3. As a Recruiter or Company Admin, I want to load an existing template for editing.

- **Tasks:**
 - Develop functionality to fetch and display saved templates in the designer.
 - Allow users to modify templates only if they have the required permissions.

- Test and debug template loading and editing to ensure seamless user experience.

4. As a System Admin, I want to ensure secure operations while using the PDF Designer.

- **Tasks:**

- Implement validation checks for all inputs to prevent security vulnerabilities.
- Log user actions related to template design for auditing purposes.
- Conduct security testing to identify and fix potential risks .

Deliverables:

- A fully functional PDF Designer interface with essential features like drag-and-drop, text/image handling, and layout customization.
- A secure backend system for storing and retrieving templates.
- Validation mechanisms to ensure security and data integrity.

4.3.2 Analysis of the Fifth Sprint

In this sprint, the focus is on developing a Feature-Rich and User-Friendly PDF Designer that empowers users (recruiters and company admins) to create and customize professional documents, particularly CV templates. This sprint emphasizes an intuitive design experience, ensuring that users can easily interact with a wide range of customization tools, including text, images, and layout adjustments.

The PDF Designer provides essential functionalities, such as drag-and-drop components, real-time preview, and the ability to save and load templates. Additionally, the system is designed to handle various permissions, ensuring that only authorized users can modify or remove specific templates.

Figure 4.9 illustrates the Sequence Diagram for this sprint.

Table 3.2 below describes one of the core use cases: Designing a New PDF Template.

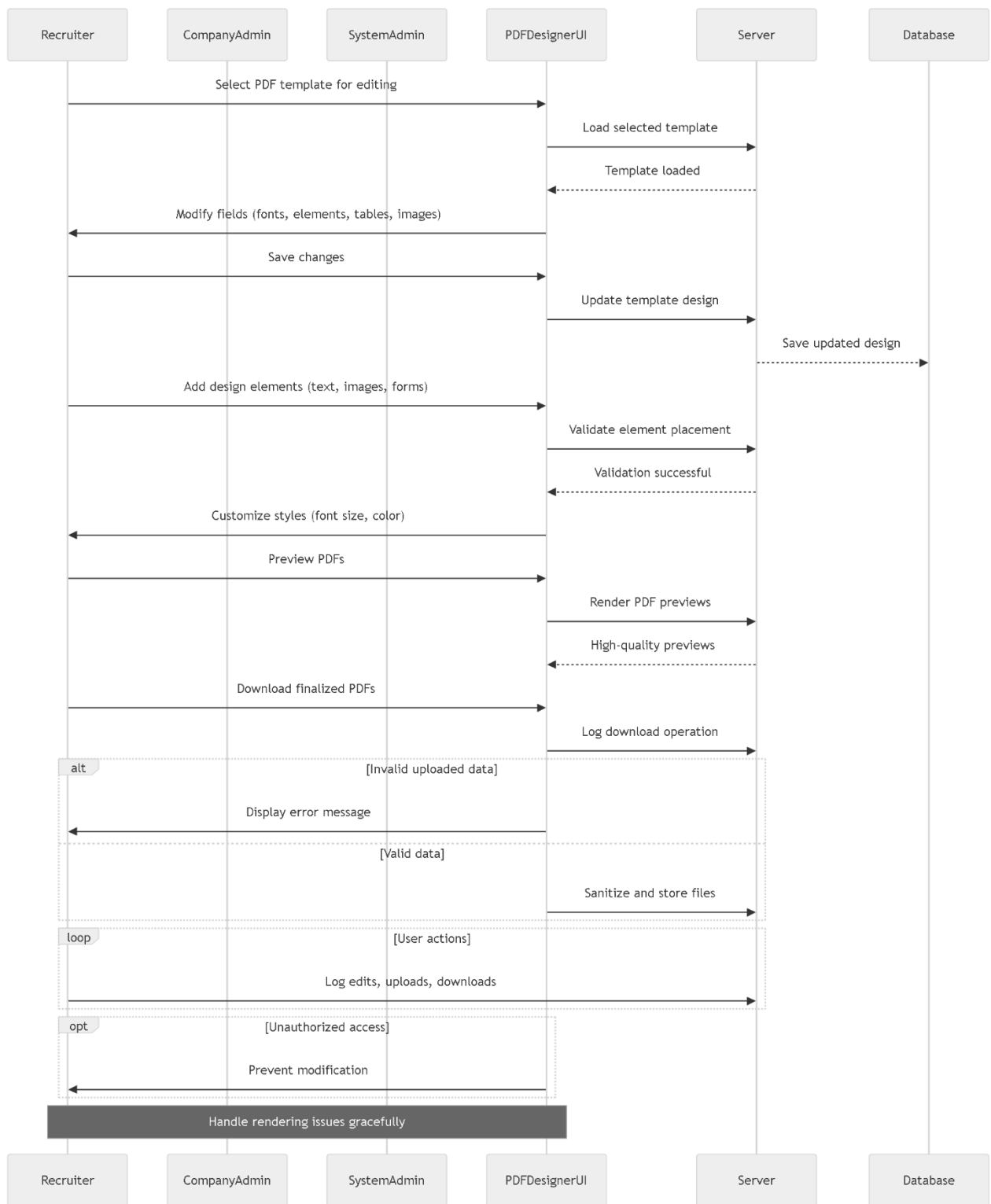


Figure 4.9 –Sequence Diagram of “A Feature-Rich and User-Friendly PDF Designer”.

Table 4.2 – Description of the " A Feature-Rich and User-Friendly PDF Designer " use case.

Use case	A Feature-Rich and User-Friendly PDF Designer
Actor	Recruiter, Company Admin
Pre-condition	The user is logged into the system and has the necessary permissions to access and modify PDF templates.
Nominal scenario	1. The user navigates to the PDF Designer interface. 2. The system loads the selected template into the designer. 3. The PDF Designer displays the template with customization options, including adding or modifying fields, text, images, forms, tables, or signatures. 4. The user modifies the template fields, customizes styles (font size, colors, alignments), or adds new elements. 5. The user previews the changes in real-time. 6. The user saves the modifications, and the system validates the changes. 7. The backend updates the modified template and ensures it is securely stored. 8. The user downloads the final PDF template or exits the designer. 9.
Exceptional scenario	1. If the uploaded data or file is invalid, the system displays an error message specifying the issue.
	2. If unauthorized users attempt to modify templates, the system denies access and logs the attempt.
	3. If there are rendering issues during the preview, the system notifies the user and provides an option to retry or contact support. The PDF template is successfully customized and saved, adhering to the user's specifications and permissions.

3.3.3 Implementation of The Fourth Sprint

This sprint involves implementing a dynamic PDF Designer with advanced customization capabilities. The tool integrates basic elements like text, images, forms, tables, and signatures while supporting bulk PDF generation and field customization. The focus is on creating an intuitive, flexible, and efficient design interface for users.

Generic fields can be injected into the template, allowing users to easily design their documents

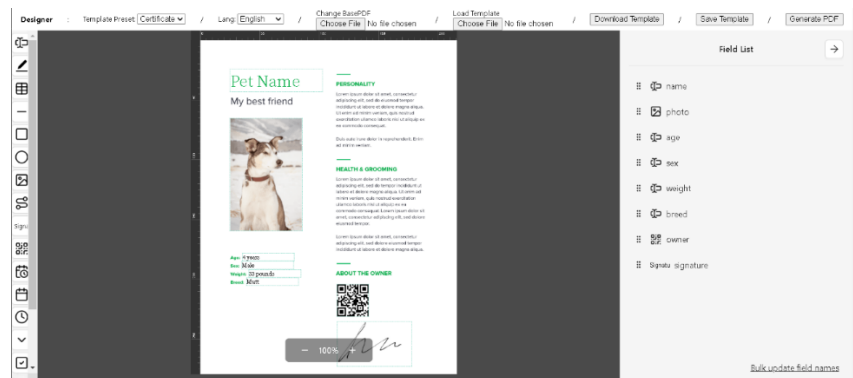


Figure 4.10 –CV Designer.

As shown in the next two figures, each injected field is fully customizable.

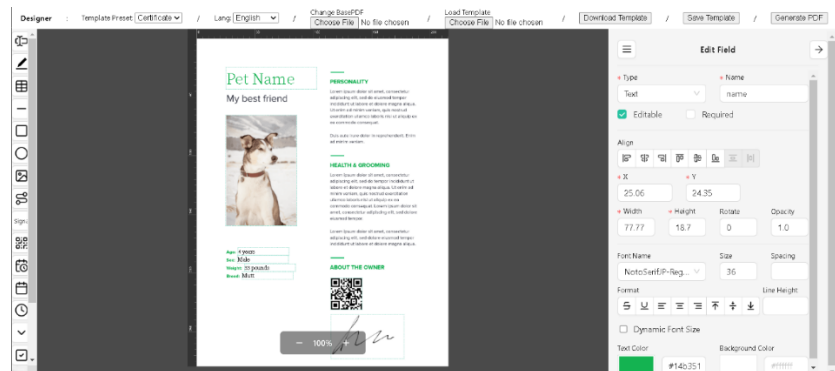


Figure 4.11 –Text customization options.

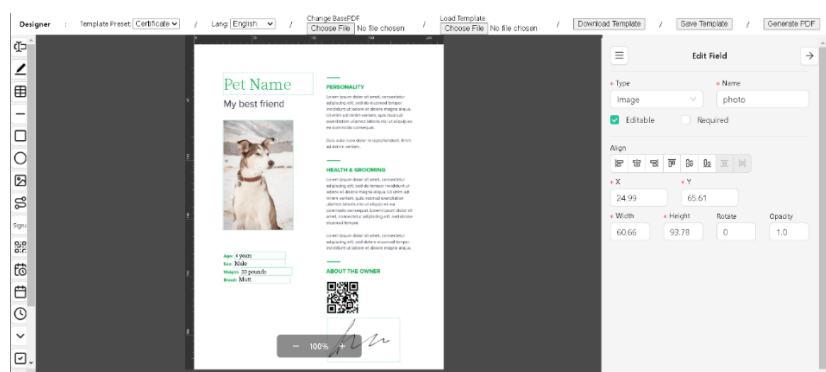


Figure 4.12 –Image customization options.

4.4 Sixth Sprint : " Custom plugins for the designer, tailored specifically for CV creation."

In this sprint, we focus on developing a highly interactive and feature-rich PDF Designer. This tool empowers recruiters and company admins to create and customize professional PDF templates with ease. The designer provides an intuitive interface, enabling users to add, modify, and style various elements such as text, images, forms, tables, and signatures. Users can also bulk-generate PDFs and customize every field, ensuring tailored documents for diverse needs.

This sprint aims to enhance user productivity and flexibility by delivering a robust and user-friendly PDF customization platform.

4.4.1 Backlog of Sixth Sprint

1. As a Recruiter or Company Admin, I want to use pre-designed CV sections (e.g., Experience, Education, General Information) as editor plugins to automatically inject them into my CV design.

- **Tasks:**

- Develop a system for pre-designed CV sections (e.g., Experience, Education, Skills, etc.) that users can inject into their CV templates.
- Provide functionality for users to customize these injected sections (e.g., adding or removing fields, changing field types).
- Ensure the CV sections are editable, and users can easily modify them to fit their needs.
- Test and debug the injection and customization process to ensure smooth functionality.

2. As a Recruiter or Company Admin, I want to customize the content of pre-designed sections to match specific needs and requirements.

- **Tasks:**

- Implement customization options for the pre-designed CV sections (e.g., text fields, date pickers, and dropdowns).
- Allow users to modify the layout and design of each section (e.g., reorder, add/remove fields).
- Enable real-time preview and feedback on customization changes.
- Test and debug customization options to ensure flexibility without breaking the design.

3. As a Recruiter or Company Admin, I want to save customized CV sections as templates for future use to improve efficiency.

- **Tasks:**

- Enable users to save customized CV sections as templates for future projects.

- Provide functionality for organizing saved templates in a user-friendly way (e.g., categorization or tagging).
- Allow users to import and export templates for sharing and reuse.
- Test and debug the template-saving and organizing functionality to ensure smooth access.

4. As a Recruiter or Company Admin, I want to integrate customized CV sections seamlessly into the overall CV layout in the PDF Designer.

- **Tasks:**

- Ensure that the custom plugins (pre-designed sections) integrate smoothly into the PDF Designer interface.
- Implement an intuitive drag-and-drop functionality for inserting and adjusting the position of sections in the CV layout.
- Allow users to apply styling, adjust alignment, and personalize content within the sections directly in the PDF Designer.
- Test and debug the integration to ensure the final output matches the design expectations.

5. As a Recruiter or Company Admin, I want to preview and validate my CV template with injected sections before finalizing it to ensure accuracy and completeness.

- **Tasks:**

- Implement a live preview feature to display the CV template with the injected sections and customized content.
- Provide a validation feature that checks for missing or invalid data in the sections.
- Allow users to test and finalize the layout and content before generating the final PDF.
- Test and debug the preview and validation process to ensure it works as expected.

Deliverables:

- A library of pre-designed CV sections (e.g., Experience, Education, Skills) that users can inject into their designs.
- A user interface for customizing and editing the injected sections.
- Functionality for saving and organizing customized sections as templates.
- Seamless integration of custom sections within the PDF Designer.
- A real-time preview and validation system for CV templates.

4.4.2 Analysis of the Sixth Sprint

In this sprint, the focus is on enhancing the PDF Designer with pre-designed, customizable CV sections that users (recruiters and company admins) can easily inject into their templates. This feature-rich sprint aims to empower users to quickly build professional

CVs by adding pre-designed sections like Experience, Education, and Skills, while also offering customization options for content, layout, and styling.

The primary objective is to provide users with a seamless experience for integrating these sections into their designs and making adjustments to suit their specific needs. Users will be able to save these customized sections as templates for future use and ensure that the final document aligns perfectly with their requirements. The system also supports various permissions, ensuring that only authorized users can modify or remove templates and sections.

Figure 4.3 illustrates the Sequence Diagram for this sprint.

Table 3.3 describes one of the core use cases: Injecting and Customizing Pre-designed CV Sections into a Template.

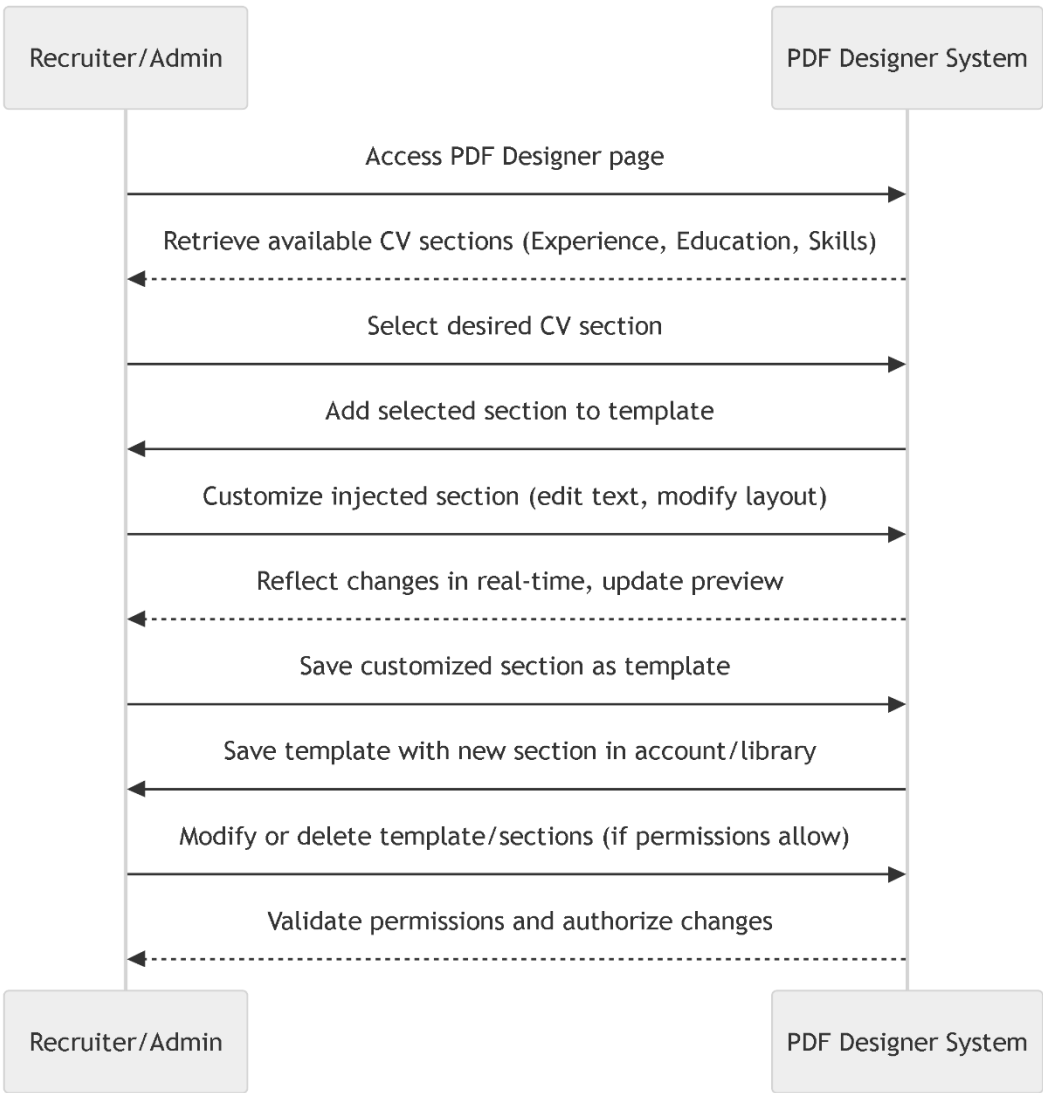


Figure 4.13 –Sequence Diagram of “Custom plugins for the designer, tailored specifically for CV creation.

Table 4.3 – Description of the " Custom plugins for the designer, tailored specifically for CV creation." use case.

Use case	Custom plugins for the designer, tailored specifically for CV creation.
Actor	Recruiter, Company Admin
Pre-condition	- The user (recruiter or company admin) is logged into the system. - The user has access to the PDF Designer tool.
Nominal scenario	1. The user navigates to the PDF Designer interface. 2. The system loads the available pre-designed CV sections (Experience, Education, Skills, etc.). 3. The user selects a pre-designed CV section (e.g., Experience). 4. The system injects the selected section into the template at the designated location. 5. The user customizes the injected section (e.g., edits text, modifies layout, adds images, etc.). 6. The system updates the preview in real-time to reflect the customizations. 7. The user clicks the "Save" button to save the customized template. 8. The system stores the customized template in the user's account or template library. 9. The user may choose to modify or delete sections if they have the necessary permissions. 10. The system ensures that only users with the required permissions (recruiter, company admin) can modify or delete certain templates.
Exceptional scenario	If the user tries to inject a section with invalid data or incorrect formatting, the system will display an error message, allowing the user to correct the issues before proceeding.
	If the user does not have the correct permissions to modify or delete a section, the system will display a permissions error and prevent the action from being performed.

3.4.3 Implementation of The Sixth Sprint

This sprint focuses on the implementation of a robust system for injecting and customizing pre-designed CV sections within the PDF Designer. The tool provides users with pre-built sections (such as Experience, Education, Skills, and more) that they can easily drag and drop into their templates. Users have the ability to fully customize these sections, adjusting text, layout, images, and more, to create personalized and professional CV templates. The system ensures smooth integration of these elements, with real-time previews and the option to save or modify templates. It also includes permission management to ensure that only authorized users (recruiters and company admins) can make changes to certain templates.

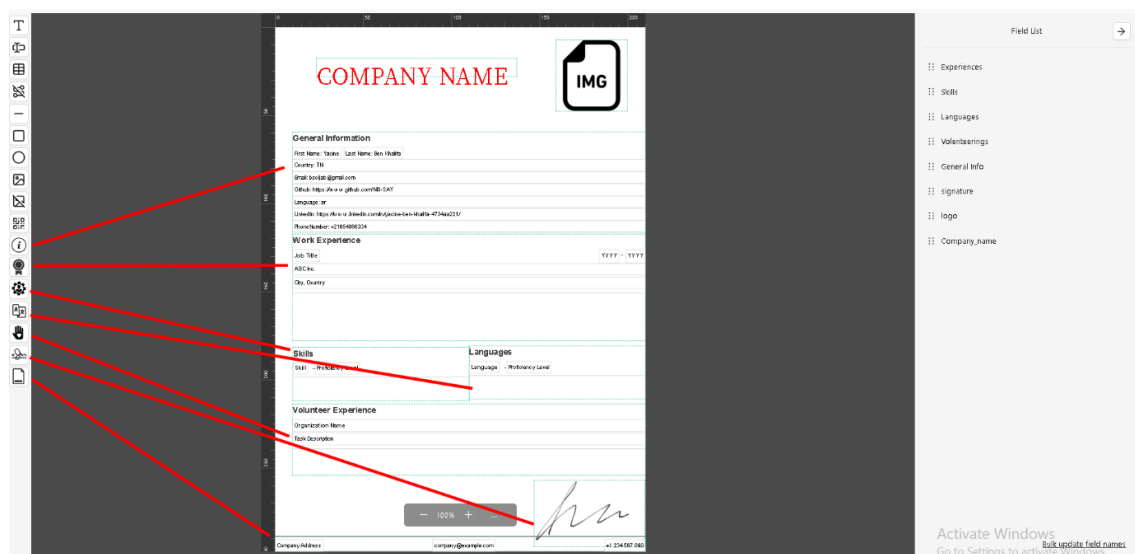


Figure 4.14 –Custom plugins integration in the designer.



Figure 4.15 –Fully customizable sections.

4.5 Conclusion

Using screenshots of the interfaces, this chapter presented the implementation of the second release with all its sprints and their modeling.

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General Conclusion

The project was developed as an end-of-studies initiative, aimed at enhancing the "HireCue" e-recruitment platform developed by Teligencia. The goal was to create a more efficient and user-friendly experience for both candidates and recruiters. The project was split into two major releases, each focusing on different functionalities to meet the needs of both groups.

Release 1 (For Candidates):

This release focused on improving the candidate's experience during the onboarding and profile management process.

- **Sprint 1: LinkedIn Data Extraction and Retrieval**
This sprint introduced a feature that enables candidates to import their professional data (such as work experience, skills, and education) directly from their LinkedIn profiles. This feature streamlined the onboarding process, saving time and reducing manual data entry.
- **Sprint 2: AI-Driven Onboarding Form**
Building on the LinkedIn data extraction, an AI-driven onboarding form was developed. This form automatically populated certain fields based on the data extracted from LinkedIn, while also allowing candidates to customize and complete their profile information. It aimed to make the profile completion process smoother and more accurate.
- **Sprint 3: Profile Management Page with All Necessary Settings**
This sprint developed a dedicated profile management page for candidates. It allowed users to view and update their personal information, including contact details, work experience, preferences, and job aspirations. The user interface was designed to be intuitive, ensuring that candidates could easily manage and update their profile.

Release 2 (For Recruiters):

This release focused on features that empowered recruiters to efficiently manage and design professional CVs.

- **Sprint 4: Highly Interactive Template Library Page**
This sprint introduced an interactive template library, allowing recruiters to browse through a variety of pre-designed CV templates. The system allowed users to select, preview, and manage templates for easy customization.
- **Sprint 5: Feature-Rich and User-Friendly PDF Designer**
A key feature of this sprint was the development of a powerful PDF designer tool. Recruiters could now design professional CVs using drag-and-drop components, real-time previews, and adjustments to various elements such as text, images, tables, and more. The tool was designed to provide a seamless, intuitive user experience.

- **Sprint 6: Custom Plugins for CV Creation**

This sprint enabled the creation of custom plugins tailored for CV generation. These plugins allowed recruiters to automatically insert pre-designed sections like work experience, education, and personal information into the CV design. It provided flexibility for further customization, making the design process faster and more efficient.

The project was developed following the Scrum methodology, with each sprint carefully planned and executed to ensure the development of high-quality features. The use of modern technologies ensured that the platform was scalable and secure, offering a seamless experience for both candidates and recruiters. Additionally, permission management ensured that only authorized users could modify certain templates or profile details.

Future Perspectives:

Moving forward, the platform could benefit from the integration of AI tools that automatically generate new templates or suggest improvements for existing templates. Additionally, adding multi-language support for CV creation and enhancing the platform with real-time collaboration features could improve user experience further. Lastly, introducing advanced document features, such as export options and job-specific formatting, could offer even more flexibility to recruiters and candidates.

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